Measure Theoretic Probability Theory

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Abstract

This project explores the branch of analysis known as measure theory and its profound consequences in probability theory. This well-structured framework enables significant advancements by unifying discrete and continuous cases, allowing the use of tools such as expectation and probability density functions in a seamless manner across both settings. We begin by laying the foundational concepts, building both intuition and technical skill, then delve into deeper analytical ideas, ultimately leading up to an exploration of the Lindeberg Central Limit Theorem.

Acknowledgements

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His excellence as a mentor and deep understanding of the subject shone not in giving me direct answers (though I often wished for shortcuts at the time), but in posing thought-provoking questions. I am truly thankful for the guidance, support, and kindness I have received.

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Preface

One of the most fundamental concepts in Euclidean geometry is that of the measure $m(E)$ of a solid body E in one or more dimensions. In one, two, and three dimensions, this measure is referred to as the length, area, or volume of E, respectively. In the classical approach to geometry, the measure of a body was often computed by partitioning it into finitely many components, applying rigid motions (such as translations or rotations) to each piece, and then reassembling them into a simpler body presumed to have the same measure. Another technique involved estimating the measure by bounding the body between inscribed and circumscribed figures—a method that dates back to Archimedes.

With the advent of analytic geometry, Euclidean geometry was reinterpreted in terms of Cartesian products $\mathbb{R}^d$ of the real line $\mathbb{R}$. Under this analytic lens, it was no longer intuitively obvious how to define the measure $m(E)$ for an arbitrary subset $E \subset \mathbb{R}^d$. We refer to the challenge of formulating a "correct" definition of measure as the *problem of measure*.

At the beginning of this work, we investigate why this problem arises and what makes measuring the real line $\mathbb{R}$ non-trivial. From there, we build a rigorous foundation for measure and integration—not merely for abstraction's sake, but as a framework that underpins essential tools and results in analysis and probability. This perspective unifies the discrete and continuous cases, enabling the same techniques—such as expectation and probability density functions—to apply across both.

We then show how probability density functions can be constructed from any distribution, whether discrete or continuous, and we conclude with an exploration of the Lindeberg Central Limit Theorem, using both analytical reasoning and numerical simulations.

1 Measure Theory

Returning to the problem of measure, example 1.0.1 illustrates that partitioning $\mathbb{R}$ into "measurable bits" is an insufficient method for assigning measure. Our goal is to develop a measure that aligns with our intuition. But first, we must ask: Are all subsets of $\mathbb{R}$ measurable?

Example 1.0.1 (Problem of Cardinality)**.** Let E_1 and E_2 be two subsets of $\mathbb{R}$, such that $E_1 = [0, 1]$ and $E_2 = [0, 2]$. Let $f : E_1 \to E_2$ be the function defined by $f(x) = 2x$. Since f is a bijection, it follows that the sets E_1 and E_2 have the same cardinality. Hence, using cardinality for measuring length contradicts our intuition.

1.1 Non-measurable Sets in $\mathbb{R}$

Before we answer if all subsets of $\mathbb{R}$ are measurable, which is not trivial to show, lets first start with some intuitive properties we want to have. To measure *length of an open interval* on $\mathbb{R}$ we want to take elements from end points of the an interval and subtract smaller element from the grater one, that is $\ell(I) = b - a$ for some $a < b \in \mathbb{R}$.

Definition 1.1.1 (Length of an Open Interval)**.** *The length $\ell(I)$ of an open interval $I \subseteq \mathbb{R}$ is defined by*

$$\ell(I) = \begin{cases} b-a & \text{if } I = (a,b) \text{ for some } a, b \in \mathbb{R} \text{ with } a < b, \\ 0 & \text{if } I = \emptyset, \\ \infty & \text{if } I = (-\infty, a) \text{ or } I = (a, \infty) \text{ for some } a \in \mathbb{R}, \\ \infty & \text{if } I = (-\infty, \infty). \end{cases}$$

Second intuitive notion that we want to have is *translation invariance*. Better put, we want to have the same measure of interval I and the same measure of $I + c$ where c is just some constant in $\mathbb{R}$ that shifts our interval I.

Definition 1.1.2 (Translation)**.** *If $t \in \mathbb{R}$ and $A \subseteq \mathbb{R}$, then the translation $t + A$ is defined by:*

$$A + t = \{a + t : a \in A\}.$$

The third property we require is *additivity of measure*. In other words, when working with disjoint sets, the measure of their union is equal to the sum of the measures of the individual sets. Additivity of measure is sometimes also referred to as σ-additivity.

Definition 1.1.3 (Additivity of Measure)**.** *Let $\ell(I)$ be the the length of an open interval and $A_i, A_j \subseteq \mathbb{R}$ be pairwise disjoint sets for every $i \neq j$. Then,*

$$\ell\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \ell(A_i) \quad \textit{where } A_i \cap A_j = \emptyset \textit{ for } i \neq j.$$

Before we dive into our first major theorem, which demonstrates that *not every subset of* $\mathbb{R}$ *is measurable*, we'll first establish the property of *monotonicity of measure*. This property states that the measure of a set is always greater than or equal to the measure of any of its subsets.

Lemma 1.1.1 (Monotonicity of Measure)**.** Let $\ell(I)$ be the length of an open interval. If $E \subseteq F$ for two $E, F \in \mathbb{R}$, then $\ell(E) \leq \ell(F)$.

Proof. Suppose E and F are subsets of $\mathbb{R}$ with $E \subseteq F$. Since E is contained in F we can write $F = E \cup (F \backslash E)$ as union of disjoints sets. Now, $\ell(F) = \ell(E \cup (F \backslash E)) = \ell(E) + \ell(F \backslash E)$. Since measure is always positive, result follows. □

We will show that it is impossible to construct a measure on $\mathbb{R}$ that is defined on all subsets of $\mathbb{R}$, assigns the length of an open interval, is translation invariant, and is σ-additive. The key step is the construction of a pathological set Ω. We will assume these four properties and use them to reach a contradiction.

Theorem 1.1.2 (Non-measurable Sets in $\mathbb{R}$)**.** There does not exist a measure on $\mathbb{R}$ satisfying all four of the following properties:

(i) $\lambda(\mathcal{P}(\mathbb{R})) \to \mathbb{R} \cup \{+\infty\}$ *(Defined on all subsets of $\mathbb{R}$)*

(ii) $\lambda((a, b)) = \ell(I)$ *(Extension of length of interval)*

(iii) $\lambda(A + t) = \lambda(A)$ *(Translation invariant)*

(iv) $\lambda\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \lambda(A_i)$ with $A_i \cap A_j = \emptyset$ *for* $i \neq j$ *(σ-Additive).*

Proof. Suppose that λ is a measure on $\mathbb{R}$ with the desired properties. Let $\sim$ be a relation on $\mathbb{R}$ defined by:

$$x \sim y \iff x - y \in \mathbb{Q}. \tag{1.1}$$

Clearly, $\sim$ satisfies the properties of symmetry, reflexivity, and transitivity. Hence, $\sim$ is an equivalence relation on $\mathbb{R}$.

$$\text{Then,}\quad [x] = \{y \in \mathbb{R} : y - x \in \mathbb{Q}\}$$

is the equivalence class of x. In other words, $[x]$ represents the set of all real numbers that can be reached from x by adding a rational number. Thus, we have decomposed $\mathbb{R}$ into disjoint equivalence classes, where each class consists of all the real numbers that can be obtained from any member of the class by adding a rational number. Since each equivalence class is countable and the union of all of them is $\mathbb{R}$, there have to be uncountably many of such equivalence classes[1].

Now, let $\Lambda = \mathbb{R}/\sim$ be the set of all such equivalence classes in $\mathbb{R}$, which partition the real numbers into disjoint sets based on the equivalence relation $\sim$. We will use the Axiom of Choice to construct a new set Ω by selecting one and only one point from each equivalence class in Λ (Figure 1.1). We can always choose representatives from equivalence classes that belong to the unit interval $(0, 1]$, so Ω is contained within $(0, 1]$.[2]

$$\Omega \subseteq (0,1], \quad \alpha, \beta \in \Omega \quad \text{where} \quad \alpha \in [\alpha], \quad \beta \in [\beta], \quad [\alpha] \neq [\beta] \quad \forall\, [\alpha], [\beta] \in \Omega.$$

We will demonstrate that there is a dichotomy when Ω is shifted by p or q, where $p, q \in \mathbb{Q}$: either the resulting sets are identical, or their intersection is the empty set.

$$\begin{aligned}
&\text{Assume } (\Omega + q) \cap (\Omega + p) \neq \emptyset \ \text{ where } \ p, q \in \mathbb{Q} \\
&\text{Then, } \exists\, x \text{ such that } x \in (\Omega + q) \text{ and } x \in (\Omega + p) \\
&\text{Now, } x = \alpha + q \ \text{ and } \ x = \beta + p \quad \text{where} \quad \alpha, \beta \in \Omega \\
&\implies \alpha - \beta = p - q \\
&\text{Since } p - q \in \mathbb{Q}, \quad \implies \alpha - \beta \in \mathbb{Q} \implies [\alpha] = [\beta] \quad \text{by definition.}
\end{aligned}$$

In the construction of the set Ω, we selected only one element from each equivalence class.

$$\begin{aligned}
&\text{Therefore,} \quad \alpha = \beta \quad \text{and} \quad q = p \quad \implies \quad \Omega + q = \Omega + p \\
&\text{It follows that when} \quad q \neq p \quad \implies \quad (\Omega + q) \cap (\Omega + p) = \emptyset.
\end{aligned}$$

[1] See the appendix for equivalence relations and set partitions.

[2] This class of pathological sets is also known as Vitali sets.

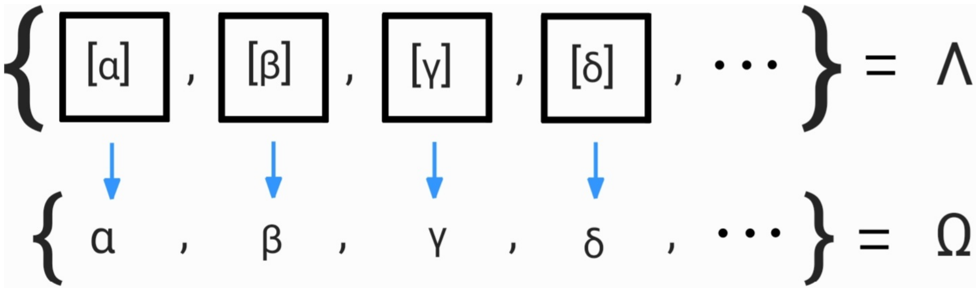

Figure 1.1: Partitioning of $\mathbb{R}$ into equivalence classes and selecting elements for set Ω
Note: elements of Λ are equivalence classes while elements of Ω are points.

Now, we will show that the measure of $\bigcup_{\substack{q\in\mathbb{Q}\\ -1<q<1}} (\Omega + q)$, is bounded above.

$$\Omega + q \subseteq (-1,2) \quad \text{for all} \quad q \in (-1,1)\cap\mathbb{Q} \quad \text{since} \quad \Omega \subseteq (0,1) \quad \text{and} \quad -1 < q < 1.$$

Since $\Omega + q$ is contained in (-1,2) for every q then the union is also contained in (-1,2).

$$\text{Hence,} \quad \bigcup_{\substack{q\in\mathbb{Q}\\ -1<q<1}} (\Omega + q) \subseteq (-1,2).$$

Property iv) tells us that the sum of measures of disjoint sets is equal to measure of union of those sets. Monotonicity property tells us that the measure of a set is greater than or equal to the measure of its subsets. Property ii) tells us that the measure of the interval $(-1,2)$ is 3. Therefore we have:

$$\sum_{\substack{q\in\mathbb{Q}\\ -1<q<1}} \lambda(\Omega + q) \overset{\text{by iv)}}{=} \lambda\Big(\bigcup_{\substack{q\in\mathbb{Q}\\ -1<q<1}} (\Omega + q)\Big) \overset{\text{by mon.}}{\leq} \lambda(-1,2) \overset{\text{by ii)}}{=} 3. \tag{1.2}$$

Finally, property iii) tells us that measure is translation invariant. Now, we are summing an infinite number of elements, so in order for to satisfy the inequality in line (1.2), it is necessary for the measure to be zero.

$$\text{Now,} \sum_{\substack{q\in\mathbb{Q}\\ -1<q<1}} \lambda(\Omega + q) \overset{\text{by iii)}}{=} \sum_{\substack{q\in\mathbb{Q}\\ -1<q<1}} \lambda(\Omega) \leq 3.$$

$$\therefore \qquad \lambda(\Omega) = 0. \tag{1.3}$$

On the other hand, we want to show the measure of $\bigcup_{\substack{q\in\mathbb{Q} \\ -1<q<1}} (\Omega + q)$ is bounded below which will give us a desired contradiction with just proven result (1.3).

$$\text{Let } x \in (0,1), \quad \text{then } \ \exists\, \alpha \in [x] \cap \Omega, \quad \alpha \in (0,1] \ \text{ since } \ \Omega \subseteq (0,1].$$

Since $x, \alpha \in [x]$, we have $x \sim \alpha$, and so:

$$\begin{aligned} &x - \alpha \overset{\text{by (1.1)}}{=} q \in \mathbb{Q} \ \text{ where } \ -1 < q < 1 \\ &\implies x = \alpha + q \in \Omega + q. \end{aligned}$$

Since x was arbitrary, then for all $x \in (0,1)$ we have that $x \in \Omega + q$ for some $q \in \mathbb{Q} \cap (-1,1)$. Therefore:

$$(0,1) \subseteq \bigcup_{\substack{q\in\mathbb{Q} \\ -1<q<1}} (\Omega + q).$$

Now, taking length of interval property we get lower bound of 1.

$$\begin{aligned} 1 &= \lambda(0,1) && \text{by ii)} \\ &\leq \lambda\Big(\bigcup_{\substack{q\in\mathbb{Q} \\ -1<q<1}} (\Omega + q)\Big) && \text{by monotonicity} \\ &= \sum_{\substack{q\in\mathbb{Q} \\ -1<q<1}} \lambda(\Omega + q) && \text{by iv)} \\ &= \sum_{\substack{q\in\mathbb{Q} \\ -1<q<1}} \lambda(\Omega) && \text{by iii)} \\ &\text{and so} \quad \lambda(\Omega) \neq 0. \end{aligned} \tag{1.4}$$

Line (1.3) contradicts (1.4), and we conclude that it is not possible to define a measure for Ω. Therefore, it is not possible to have a measure defined on all subsets of $\mathbb{R}$, that is an extension of length of interval, that is translation invariant and that is σ-additive. □

Theorem 1.1.2 suggests that, in order to construct a well-defined measure, we must relax al least one of the properties. Intuitively, it would be problematic to lose translation invariance. We cannot forgo the additivity of measures, as it will be essential later when proving theorems related to limits. Moreover, we want to extend the notion of length we defined on open intervals. To achieve this, the property we choose to relax is the set of measurable sets themselves. Specifically, we will restrict our subsets of $\mathbb{R}$ from the full power set to a class of sets that we will define as measurable in the next subchapter.

1.2 Measurable Sets and Measures

We will examine measurable sets, which are key objects in probability and measure theory. A measurable set is one for which we can assign a well-defined probability or measure, using the concept of a σ-field. This framework ensures consistency when assigning probabilities to events in random experiments.

Definition 1.2.1 ($\sigma - field$)**.** *: Let X be a set. A $\sigma - field$ on X is collection $\mathcal{A} \subseteq \mathcal{P}(\mathbb{X})$ which obeys the following properties:*

$$\begin{aligned} &(i) \quad \emptyset \in \mathcal{A} \quad \textit{and} \quad X \in \mathcal{A}. \\ &(ii) \quad \textit{If } A \in \mathcal{A}, \textit{ then } A^c \in \mathcal{A}. \\ &(iii) \quad \textit{If } A_1, A_2, A_3, \dots \in \mathcal{A}, \quad \textit{then } \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}. \end{aligned}$$

We refer to pair $(\mathbb{X},\ \mathcal{A})$ of a set $\mathbb{X}$ together with $\sigma - field$ on that set as a measurable space. Note that this definition implies that a σ-field also contains countable intersections of sets. Indeed, if $\{A_i\}_{i=1}^{\infty} \in \mathcal{F}$, then

$$\left(\bigcap_{i=1}^{\infty} A_i\right)^c = \bigcup_{i=1}^{\infty} A_i^c \Rightarrow \bigcap_{i=1}^{\infty} A_i \in \mathcal{F}.$$

This follows from De Morgan's laws. Two sets such that $\mathcal{A} \subseteq \mathcal{P}(\mathbb{X})$ that form $\sigma - fields$ are $\mathcal{A} = \{\emptyset, \mathbb{X}\}$ and $\mathcal{A} = \mathcal{P}(\mathbb{X})$. For more interesting example we will introduce following definition.

Definition 1.2.2 (Smallest σ-field on Set $\mathcal{M}$)**.** *For $\mathcal{M} \subseteq \mathcal{P}(\mathbb{X})$ there is a smallest σ-field that contains $\mathcal{M}$:*

$$\sigma(\mathcal{M}) = \bigcap_{\mathcal{M} \subseteq \mathcal{A}} \mathcal{A} \quad \text{where } \mathcal{A} \text{ are all } \sigma - \text{fields containing } \mathcal{A}.$$

We can visualize an example 1.2.1 of a $\sigma(\mathcal{M})$ with figure 1.2. We began with space $\mathbb{X} = \{1, 2, 3, 4\}$ and set containing two elements $\mathcal{M} = \{\{1, 2\}, \{1, 2, 3\}\}$ and applied the properties of $\sigma - field$ to generate *a measurable set* $\sigma(\mathcal{M})$ which is the smallest σ-field containing $\mathcal{M}$. Sometimes $\sigma - field$ is also refered to as $\sigma - algebra$.

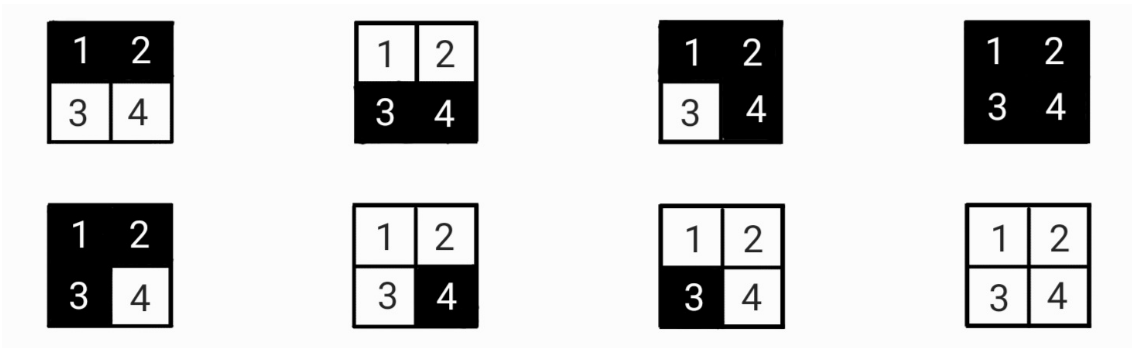

Figure 1.2: Example of $\sigma(\mathcal{M})$

Example 1.2.1 ($\sigma(\mathcal{M})$)**.** Let space $\mathbb{X} = \{1, 2, 3, 4\}$ and set $\mathcal{M} = \{\{1, 2\}, \{1, 2, 3\}\}$. To construct σ-algebra on $\mathcal{M}$ we need to add $\emptyset$ and X, as well as make sure $\mathcal{M}$ is closed under countable union and it is closed under complementation.
Now, $\{1, 2\}^c = \{3, 4\}$, $\{1, 2, 3\}^c = \{4\}$, $\{1, 2\} \cup \{4\} = \{1, 2, 4\}$, $\{1, 2, 4\}^c = \{3\}$, $\{1, 2\} \cup \{3, 4\} = \{X\}$, $\{X\}^c = \{\emptyset\}$.
Finally, we have $\sigma(\mathcal{M}) = \{\{\emptyset\}, \{X\}, \{1, 2\}, \{1, 2, 3\}, \{3, 4\}, \{4\}, \{1, 2, 4\}, \{3\}\}$.

Notice that our $\sigma(\mathcal{M})$ did not generate the sets $\{1\}$ or $\{2\}$. Before offering a more intuitive explanation, we will first introduce the definition of a special σ-field, specifically the Borel σ-field on $\mathbb{R}$, denoted $\mathcal{B}(\mathbb{R})$, where the concept of open sets must be well-defined.

Definition 1.2.3 (Borel σ-field)**.** *The Borel σ-field on $\mathbb{R}$, denoted $\mathcal{B}[\mathbb{R}]$, is the smallest σ-field that contains every open subset of $\mathbb{R}$. That is,*

$$\mathcal{B}[\mathbb{R}] = \sigma(\{O \subseteq \mathbb{R} : O \textit{ is open}\}).$$

It is important to note that $\{O \subseteq \mathbb{R} : O \text{ is open}\}$ is the same set as $\{(a, b) \subseteq \mathbb{R} : \ a < b\}$. This is because every open subset of the real line is a finite or countable union of open intervals. Then, we may also think of $\mathcal{B}[\mathbb{R}]$ as the smallest σ-field containing all open intervals (a, b) in $\mathbb{R}$.

Another important concept we will introduce is *measure*. A measure is a non-negative function that assigns a value of zero to the empty set and satisfies the property that the measure of the union of disjoint sets equals the sum of the measures of the individual disjoint sets.

Definition 1.2.4 (Measure)**.** *: A map $\mu : \mathcal{A} \to [0, \infty]$ is called measure if it satisfies:*

(i) $\mu(\emptyset) = 0$

(ii) $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{1}^{\infty} \mu(A_i) \quad with \quad A_i \bigcap A_j = \emptyset,\ i \neq j,\ \forall \in \mathcal{A} \quad (\sigma - Additive).$

There are several important special cases of measures μ that are of particular interest. For a *measurable space* $(\Omega, \mathcal{F}, \mu)$, we say that μ is a *probability measure* if $\mu(\Omega) = 1$. In such cases, the triplet $(\Omega, \mathcal{F}, \mu)$ is referred to as a *probability space*. Typically, the measure μ is denoted by P in probability theory.

An example of a probability measure is the uniform distribution on the interval $[0, 1]$, where the probability of any subinterval is proportional to its length. That is,

$$P(A) = \int_A 1\, dx \quad \text{for any measurable set } A \subseteq [0, 1],$$

where $P([0, 1]) = 1$. This property ensures that the total probability of the entire interval $[0, 1]$ is equal to 1, which is a fundamental characteristic of any probability measure.

A measure μ is called a *finite measure* if $\mu(\Omega) < \infty$. For example, consider any finite interval in $\mathbb{R}$, to denote space $\mathbb{X}$. Let measure ℓ on $\mathbb{X}$ be lenght of interval as defined before. This measure is finite because it assigns a finite value to the whole space $\mathbb{X}$.

A measure μ is called a σ*-finite measure* if the set $\mathbb{X}$ can be covered by at most countably many measurable, disjoint sets with finite measure. Specifically, there exist sets $B_1, B_2, \dots \in \mathcal{A}$ such that $\mu(B_n) < \infty$ for all $n \in \mathbb{N}$, and the sets B_i are disjoint, i.e., $B_i \cap B_j = \varnothing$ for $i \neq j$, and their union covers $\mathbb{X}$, that is, $\bigcup_{n\in\mathbb{N}} B_n = \mathbb{X}$. An example of a σ-finite measure would be again measure ℓ we used for example of finite measure, only now, $\mathbb{X} = \mathbb{R} = \bigcup_{n\in\mathbb{Z}} [n, n + 1]$ and $\ell(I) = 1$ for each n. That is, we cover infinite set $\mathbb{R}$ with infinite countable number of finite intervals so we have σ-finite measure.

Return to the example 1.2.1. One can think of $\mathbb{X}$ as a four-sided die, and the sets $\{1, 2\}$ and $\{1, 2, 3\}$ as events with known probabilities $P_1(\{1, 2\})$ and $P_2(\{1, 2, 3\})$. We are interested in determining what other outcomes can be measured. We can find the probability

$P_3(\{3,4\}) = 1 - P_1(\{1,2\})$. Similarly, we can determine the probabilities of all other events in $\sigma(\mathcal{M})$. However, it was not specified that we are working with a fair die, so there are multiple possibilities for the probabilities $P(\{1\})$ and $P(\{2\})$. Therefore, we conclude that we are not given enough information, and hence $\{1\}, \{2\} \notin \sigma(\mathcal{M})$. In fact, such probability examples inspired the definition of $\sigma - field$.

1.3 Constructing Measures from Pre-measures

In this section, we prove what is typically referred to as the Carathéodory Extension Theorem. This fundamental result in measure theory tells us how to extend a pre-measure defined on an algebra of sets to a measure on a σ-field of sets. But first, we need to define these objects.

We will begin by defining an algebraic structure known as a *semi-ring*. This concept is crucial because the set of all semi-open intervals, which is of particular interest to us, forms the structure of a semi-ring. Note that the definitions of semi-ring, ring, and field differ between measure theory and abstract algebra. In measure theory, the focus is on set operations such as union, intersection, and relative difference, whereas in abstract algebra, the focus is on algebraic operations like addition and multiplication.

Definition 1.3.1 (Semiring)**.** *A collection of sets A of $\mathbb{X}$ is a semiring if $\emptyset \in A$ and for all $A, B \in A$, we have $A \cap B \in A$ and $B \setminus A = \bigcup_{i=1}^{n} C_i$ where $C_i \in A$ for $i = 1, \dots, n$.*

Example 1.3.1 (Set of all Semi-closed intervals)**.** We define $\mathcal{A}$ as a set of all semi-closed intervals on $\mathbb{R}$. That is,

$$\mathcal{A} = \{\text{all semi-closed intervals} : a, b \in \mathbb{R}\}.$$

To prove that $\mathcal{A}$ is a semiring, we must demonstrate that $\emptyset$ is contained in $\mathcal{A}$, that $\mathcal{A}$ is closed under intersection, and that the set difference between any two elements of $\mathcal{A}$ can be expressed using elements of $\mathcal{A}$.

We define $\emptyset$ as semi-closed interval where $a = b$, so $\emptyset \in \mathcal{A}$. To demonstrate other two properties, we consider following three cases: when the intervals have no common elements, when they partially overlap, and when one the intervals is contained within the other as shown in Figure 1.3. The intersection of intervals A and B is $A \cap B = [\max(a, c), \min(b, d))$ which is a semi-closed interval, so $A \cap B \in \mathcal{A}$.

To show, $B \setminus A$ is contained in $\mathcal{A}$, without loss of generality, we will refer to our three cases. When the intervals do not overlap $B \setminus A = B \in \mathcal{A}$. When intervals partially overlap $B \setminus A = (b, d] \in \mathcal{A}$. Lastly, when one the intervals is contained within the other $B \setminus A = (c, a] \cup (b, d]$ which is union of semi-open intervals so $B \setminus A \in \mathcal{A}$. We conclude $\mathcal{A}$ is a semiring.

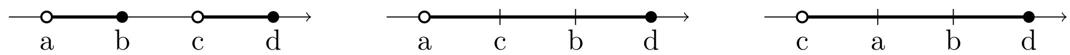

Figure 1.3: Three cases of overlapping two semi-open intervals $(a, b]$ and $(c, d]$ where $a < b$ and $c < d$.

Having shown this, we will easily demonstrate that by adding finite unions to a semiring, we obtain a new algebraic structure called a *ring*, denoted by $\mathcal{A}$. Note that every ring is a semiring, but the reverse does not always hold.

Definition 1.3.2 (Ring)**.** *A collection of sets A of $\mathbb{X}$ is a ring if $\emptyset \in A$ and for all $A, B \in A$, both $B \setminus A \in A$ and $A \cup B \in A$.*

Example 1.3.2 (Set of all Finite Unions of Semi-open intervals)**.** Let

$$\mathcal{A} = \{\text{all finite unions of semi-closed intervals} : a, b \in \mathbb{R}\}.$$

Similar to example 1.3.1, $\emptyset \in \mathcal{A}$. Union of any $A, B \in \mathcal{A}$ is a finite unions of semi-closed intervals so $A \cup B \in \mathcal{A}$. For any $A, B \in \mathcal{A}$, if $A \geq B$ then $B \setminus A = \emptyset \in A$ or $A < B$ then $B \setminus A$ is finite unions of semi-open intervals so $B \setminus A \in \mathcal{A}$. We conclude $\mathcal{A}$ is a ring.

In mathematics, it's common to understand a concept by illustrating what it is not. In the example below, we will present a field that is not a σ-field. The key distinction demonstrated here is that a field is closed under finite union, while a σ-field is closed under countable unions, which can involve infinitely many sets.

Definition 1.3.3 (Field)**.** *A ring $\mathcal{F}$ is a field if $\mathbb{X} \in \mathcal{F}$.*

Example 1.3.3 (Field that is not a σ-field)**.** If we start with any space $\mathbb{X}$ we can define a $\mathcal{F}$ on that set by

$$\mathcal{F} = \{A \subseteq \Omega : \text{either } A \text{ or } A^c \text{ is finite}\}.$$

First we will show that $\mathcal{F}$ is closed under union. Now, $\varnothing$ is finite so $\varnothing, X \in \mathcal{F}$. Let us assume some $A \in \mathcal{F}$, then either $A = (A^c)^c$ or A^c is finite, so $A^c \in A$. Let $A, B \in \mathcal{F}$. If both A and B are finite, then $A \cup B$ is also finite, hence $A \cup B \in \mathcal{F}$. Further, if both A and

B are not finite, or at least one of them is infinite (countably infinite), then since $A, B \in \mathcal{F}$, at least one of A^c or B^c is finite. Hence, $A^c \cap B^c = (A \cup B)^c$ is finite, so $A \cup B \in \mathcal{F}$.

In other to show that $\mathcal{F}$ is a field we still need to show that $B \backslash A \in \mathcal{F}$ for any $B, A \in \mathcal{F}$. When both A and B are finite, $B \backslash A$ is finite so $B \backslash A \in \mathcal{F}$. When B is infinite and A is finite, knowing B^c is finite we write $(B \backslash A)^c = (B \cap A^c)^c = B^c \cup A$ which is finite so $B \backslash A \in \mathcal{F}$. Lastly, when B and A are infinite, their complemets are finite. Hence, $B \backslash A = B \cap A^c$ is finite so $B \backslash A \in \mathcal{F}$. Thus, $\mathcal{F}$ is a field.

Now we will show and example of $\mathcal{F}$ that is a field but not σ-field. Let $X = \mathbb{Z}$. Then all singletons $\{n\}$ for $n \geq 0$ lie in $\mathcal{F}$. But the union $\bigcup_{n=0}^{\infty} \{n\} = \mathbb{N}$ and its complement, the negative whole numbers, are both infinite and hence not in $\mathcal{F}$. Now, property (iii) of σ-field necessitates $\bigcup_{n=0}^{\infty} \{n\} \in \mathcal{F}$. We conclude $\mathcal{F}$ is a field that is not a σ-field.

We will introduce a class of functions called *set functions*, which share properties with *measures*, though this class is more general. Specifically, we are interested in a special case of set functions that are countably additive and satisfy $\mu(\emptyset) = 0$, which we will refer to as *pre-measures*. The goal is to extend a pre-measure defined on a ring $\mathcal{A}$ to a measure on $\mathcal{B}(\mathbb{R})$.

Definition 1.3.4 (Set Functions)**.** *For a general set function $\mu : \mathcal{A} \to \mathbb{R}^+$ (i.e., not necessarily a measure) and $A, B \in \mathcal{A}$, we say that:*

- *μ is increasing if for $A \subseteq B$, $\mu(A) \leq \mu(B)$.*
- *μ is additive if for A, B disjoint, $\mu(A \cup B) = \mu(A) + \mu(B)$.*
- *μ is countably additive if for $\{A_i\}_{i=1}^{\infty}$ pairwise disjoint with $\bigcup_{i=1}^{\infty} A_i \in A$,*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

- *μ is countably subadditive if for $\{A_i\}_{i=1}^{\infty}$ with $\bigcup_{i=1}^{\infty} A_i \in A$,*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i).$$

The final concept we need to introduce before proving Carathéodory Extension Theorem is *outer measure*. In Figure 1.4, we show three examples of coverings of set E, each progressively tighter and providing a more accurate outer approximation. Taking the infimum over all finite and countably infinite collections of coverings gives us the outer measure.

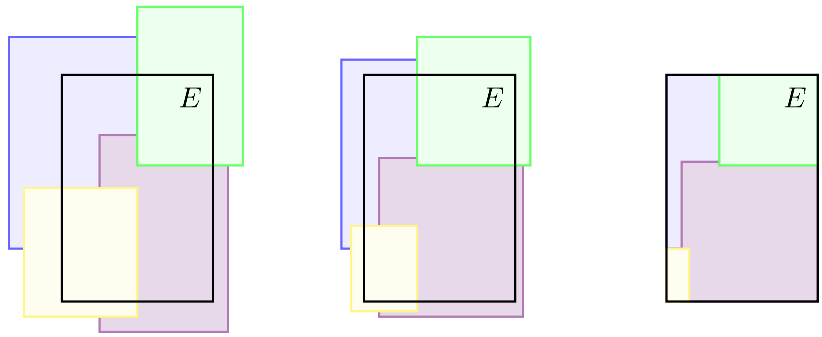

Figure 1.4: Covering of set E

Definition 1.3.5 (Outer Measure)**.** *The outer measure μ^* of a set $E \subseteq \mathbb{X}$ is defined by*

$$\mu^*(E) = \inf \left\{ \sum_{i=1}^{\infty} \mu(A_i), \quad \textit{where } \{A_i\} < \infty \textit{ and } E \subseteq \bigcup_{i=1}^{\infty} \{A_i\} \right\}.$$

We say that a set $B \subset \mathbb{X}$ is μ^*-measurable if for every $E \subset \mathbb{X}$, the outer measure μ^* satisfies additivity on $E \cap B$ and $E \cap B^c$, i.e., $\mu^*(E) = \mu^*(E \cap B) + \mu^*(E \cap B^c)$. This concept is illustrated in figure 1.5.

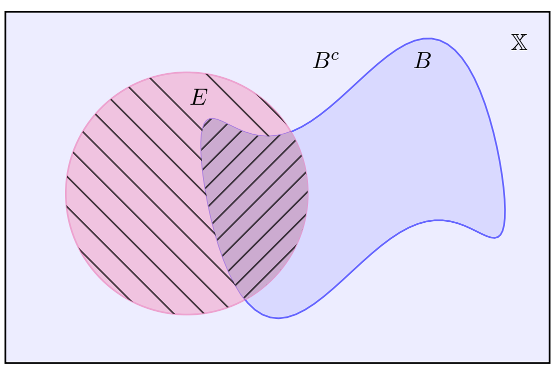

Figure 1.5: μ^* measurable sets

Definition 1.3.6 (μ^*-measurable Sets)**.** *Let μ^* be an outer measure on a set $\mathbb{X}$. A set $B \subset \mathbb{X}$ is said to be μ^*-measurable if*

$$\mu^*(E \cap B) + \mu^*(E \cap B^c) = \mu^*(E)$$

for every set $E \subset \mathbb{X}$.

Our aim is to show that μ^* is the correct way to extend μ from $\mathcal{A}$ to $\sigma(\mathcal{A})$. We also want to show that set of all μ^*-measurable sets $\mathcal{M}$ is a σ-field and that it contains $\sigma(\mathcal{A})$.

Definition 1.3.7 (Set of all μ^*-measurable Sets). *Let $\mathcal{M}$ be the set of all μ^*-measurable sets*

The Carathéodory extension theorem provides a method for extending a pre-measure defined on a ring of sets to a measure on the σ-field generated by that ring. The theorem ensures that a countably additive set function defined on a smaller collection of sets can be extended to a larger collection, maintaining the properties of additivity and consistency. This result is fundamental in measure theory, enabling the construction of measures from pre-measures, which is essential for the development of integration theory and probability.

Theorem 1.3.4 (Carathéodory Extension Theorem). *Let $\mathcal{A}$ be a ring on $\mathbb{X}$ and μ be a pre-measure. Then, μ extends to a measure on $\sigma(\mathcal{A})$.*

Proof. This proof will proceed in multiple steps. We assume that the sets $B \subseteq \mathbb{X}$ below have finite measure, that is, $\mu^*(B) < \infty$. Otherwise, the results can still be shown to trivially hold.

Step (1) We first prove a few properties of μ^*.

i $\mu^*(\emptyset) = 0$, which follows from μ being a pre-measure.

ii μ^* is *non-negative* for all $B \subseteq \mathbb{X}$, which follows from the non-negativity of μ.

iii μ^* is *monotone*. Let $B_1, B_2 \in \mathcal{A}$ and $B_1 \subseteq B_2$, then for any $\{A_i\}$ such that $B_2 \subseteq \bigcup_i A_i$, then $B_1 \subseteq \bigcup_i A_i$. Therefore, $\mu^*(B_1) \leq \mu^*(B_2)$.
In other words, for any two subsets of a ring $\mathcal{A}$ where one of the subsets is contained in the other, if we cover the larger set, we must also have covered the smaller set.

iv μ^* is *countably subadditive*. For $\{B_i\}_{i=1}^{\infty}$ and a given $\varepsilon > 0$, let $B_i \subseteq \bigcup_j A_{ij}$ for $A_{ij} \in \mathcal{A}$ such that $\sum_j \mu(A_{ij}) \leq \mu^*(B_i) + \varepsilon\ 2^{-i}$. Note, μ is more convenient to work with than μ^* and since we are working on the ring $\mathcal{A}$ we can compare μ^* with μ and reach conclusions. As $\bigcup_{i=1}^{\infty} B_i \subseteq \bigcup_{i,j} A_{ij}$ and since μ^* is monotone and μ is subadditive, we get

$$\mu^*\left(\bigcup_{i=1}^{\infty} B_i\right) \overset{\text{by mon.}}{\leq} \mu\left(\bigcup_{i,j} A_{ij}\right) \overset{\text{by subadd.}}{\leq} \sum_{i,j} \mu(A_{ij}) \leq \sum_i \mu^*(B_i) + \varepsilon.$$ [3]

As $\varepsilon > 0$ is arbitrary, this implies μ^* is countably subadditive.

Step (2) Check that μ and μ^* *coincide* on $\mathcal{A}$ for all $A \in \mathcal{A}$. That is, measuring any A gives the same value for both μ and μ^*. For any $A \in \mathcal{A}$, we immediately have that $\mu^*(A) \leq \mu(A)$ since $A \subseteq A$. For the reverse, if $A \subset \bigcup_i A_i$, then by countable subadditivity and monotonicity, we have

$$\mu(A) \overset{\text{by subadd.}}{\leq} \sum_i \mu(A \cap A_i) \overset{\text{by mon.}}{\leq} \sum_i \mu(A_i). \tag{1.5}$$

Thus, $\mu(A) \leq \mu^*(A)$ by taking the infimum of 1.5. That is to say, measure of any covering of A is greater than or equal to the measure of the set being covered. By taking the smallest such covering, we get the desired inequality. It follows that $\mu(A) = \mu^*(A)$.

Step (3) Check that $\mathcal{A} \subset \mathcal{M}$. That is to say for any $A \in \mathcal{A}$, we need to show that A is μ^*-*measurable.* Hence, for any $A \in \mathcal{A}$ and all $E \subset \Omega$, we want

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Since $\mu^*(E \cap A) + \mu^*(E \cap A^c) \geq \mu^*(E)$ by subadditivity shown in Step (1), it suffices to show that $\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(E)$. Now, for some $\varepsilon > 0$, choose $\{A_i\}$ such that $E \subseteq \bigcup_i A_i$ and $\sum_i \mu(A_i) \leq \mu^*(E) + \varepsilon$. The idea is that we are selecting a covering that is close to the infimum, and by adding some ε, we make the infimum larger than the covering.

Similar tehnique as above, since we are working on the ring $\mathcal{A}$ we can compare μ^* and μ. Furthermore, $E \cap A \subseteq \bigcup_i (A \cap A_i)$ and $E \cap A^c \subseteq \bigcup_i (A^c \cap A_i)$. Also, by countable additivity we have equality when the sets are disjoint.

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \sum_i \mu(A \cap A_i) + \sum_i \mu(A^c \cap A_i) \overset{\text{by add.}}{=} \sum_i \mu(A_i) \leq \mu^*(E) + \varepsilon,$$

[3] A neat technique often used in measure theory involves taking $\varepsilon 2^{-i}$. When summing over all i, the total converges to ε, since $\sum_{i=1}^n 2^{-i} \to 1$ as $n \to \infty$.

which shows that $A \subset \mathcal{M}$ since ε was arbitrary.

Step (4) Show that $\mathcal{M}$ is a *σ-field.* We begin by verifying that $\mathcal{M}$ is a field by checking that $\mathbb{X}$ is contained in $\mathcal{M}$. Now, $\emptyset \in M$ as $\emptyset \in \mathcal{A}$. For $\mathbb{X}$ and any $E \subseteq \mathbb{X}$,

$$\mu^*(E \cap \mathbb{X}) + \mu^*(E \cap \emptyset) = \mu^*(E),$$

and hence $\mathbb{X} \in \mathcal{M}$. Now, take any $B \in M$ on any $E \subseteq \mathbb{X}$ and using $B = \left(B^c\right)^c$, it follows directly that $\mathcal{M}$ is closed under complementation.

$$\mu^*(E \cap B) + \mu^*(E \cap B^c) \Longrightarrow \mu^*(E \cap B^c) + \mu^*(E \cap (B^c)^c) = \mu^*(E)$$

To extend from a field to a σ-field, we need to show that for a countable pairwise disjoint collection $\{B_i\}$ in $\mathcal{M}$, that $\bigcup_i B_i \in \mathcal{M}$. Let $B = \bigcup_{i=1}^{\infty} B_i$. Proceeding, once again, from the definition,

$$\begin{aligned}
\mu^*(E) &= \mu^*(E \cap B_1) + \mu^*(E \cap B_1^c) \\
&= \mu^*(E \cap B_1) + \mu^*(E \cap B_2) + \mu^*(E \cap B_1^c \cap B_2^c) \\
&= \sum_{i=1}^{n} \mu^*(E \cap B_i) + \mu^*(E \cap \Big\{ \bigcap_{i=1}^{n} B_i^c \Big\}).
\end{aligned}$$

By monotonicity, subadditivity, and taking $n \to \infty$, we get

$$\mu^*(E) \geq \sum_{i=1}^{\infty} \mu^*(E \cap B_i) + \mu^*(E \cap B^c) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c) \geq \mu^*(E).$$

Thus, M is closed under countable unions. Finally, choosing $E = B$ above, we have

$$\mu^*(B) = \sum_{i=1}^{\infty} \mu^*(B \cap B_i)$$

and thus μ^* is countably additive.

Step (5) The conclusion. What we have from all of the above is that μ^* is a set function on the power set $\mathcal{P}(\Omega)$, but it is also a measure in $\mathcal{M} \subset \mathcal{P}(\Omega)$. Furthermore, since $A \subset \mathcal{M}$ and $\mathcal{M}$ is a σ-field, we have that $\sigma(A) \subseteq M$. Lastly, since μ^* is a measure on $\mathcal{M}$, it is also a measure on any sub-σ-field. Hence, it is a measure on $\sigma(A)$. $\square$

Given such an extension as above, we wish to know whether or not it is *unique.* That is, if

μ_1 and μ_2 are measures on $\sigma(A)$ and if $\mu_1(A) = \mu_2(A)$ for any $A \in \mathcal{A}$, then is it also true that $\mu_1(B) = \mu_2(B)$ for any $B \in \sigma(A)$? That is, if two pre-measures μ_1 and μ_2 agree on the ring $\mathcal{A}$, do they also agree on the σ-algebra generated by $\mathcal{A}$? In that case, would μ_1 and μ_2 be the same measure since they coincide everywhere.

The answer is yes: our extension is *unique.* The proof is omitted due to page limitations. Similar to Carathéodory Extension Theorem, the proof primarily involves routine verifications of various properties and does not substantially contribute to the insights of this work. A detailed proof can be found in [Dudley, 2018].

1.4 Completing a Measure

In this section, we want to complete a measure. That is, if some arbitrary set E only differs from a measurable set A on a set of measure zero, then we wish to assign the same measure to both sets. However, our σ-field may not contain all sets that should be measure zero. This is made more precise below.

Definition 1.4.1 (Symmetric Difference)**.** *For two sets A, B, the symmetric difference is*

$$A \Delta B = (A \setminus B) \cap (B \setminus A).$$

For a measure space $(\mathbb{X}, \mathcal{F}, \mu)$ with $\mathcal{F} \subset \mathcal{P}(X)$, we can define the outer measure μ^* on $P(X)$ as above:

$$\mu^*(B) = \inf\{\mu(A) : B \subset A\} \quad \text{for any } B \in \mathcal{P}(X).$$

Then, we can define the set of μ-null sets to be $N_\mu \subset \mathcal{P}(X)$, where $N_\mu = \{N \subset \mathbb{X} : \mu^*(N) = 0\}$. A measure space $(\mathbb{X}, \mathcal{F}, \mu)$ is complete if $N_\mu \subset \mathcal{F}$. Also, N_μ is a ring.

The completion of σ-field $\mathcal{F}$ with respect to μ is $\mathcal{F} \vee N_\mu = \{A \cup N : A \in \mathcal{F}, N \in N_\mu\}$. In [Dudley, 2018, Proposition 3.3.2], it is proven that this completion is equal to $\{B \subset \mathbb{X} : \exists A \in \mathcal{F}$ such that $A \Delta B \in N_\mu\}$ and that this set is the smallest σ-field that contains both $\mathcal{F}$ and N_μ. Thus, we can define the completed measure space to be $(\mathbb{X}, \mathcal{F} \vee N_\mu, \bar{\mu})$ where $(\bar{\mu}(A \cup N) = \mu(A) \quad \text{for } A \in \mathcal{F}$ and $N \in N_\mu$.

1.5 Lebesgue Measure

The Carathéodory Extension Theorem and allow us to construct Lebesgue measure denoted λ, which is a central tool of measure theory. It is also of interest that $\lambda((a, b]) = b - a$ and that λ is the only such measure with this property. A standard way to construct Lebesgue measure on $(0, 1]$ or on $\mathbb{R}$ is to begin with the set of finite unions of half-open intervals $(a, b]$. Both cases are of interest as the Lebesgue measure on the unit interval corresponds to the uniform distribution and Lebesgue on $\mathbb{R}$ is necessary for defining probability density functions.

Symbol λ is often used for both the pre-measure assigning lengths to unions of intervals from $\mathcal{A}$ and the outer measure on the Borel σ-field $\mathcal{B}$. We will also denote M_λ to be the set of all Lebesgue measurable sets. It's worth noting that $\mathcal{B} \subset M_\lambda$. In fact, M_λ is the completion of $\mathcal{B}$ with N_λ. In the next section, we will discuss sets that are Lebesgue measurable but do not belong to $\mathcal{B}$.

1.6 Non-empty Sets with Measure Zero

First, we define what it means to have *measure zero*, a concept that will be useful later when discussing integrals. Then, we provide a few examples.

Definition 1.6.1 (Zero Measure)**.** *We say that a set $E \subset \mathbb{R}$ is null or has measure zero if $m^*(E) = 0$.*

Lemma 1.6.1 (Lenght of Countable Subset in $\mathbb{R}$)**.** *Every countable subset of R has measure zero.*

Proof. First we will show that the singleton $\{x\}$ satisfies $m^*(\{x\}) = 0$ for any $x \in \mathbb{R}$.

Let $\{x\} \subseteq \mathbb{R}$ and let $\varepsilon > 0$. We can cover x with open intervals of length $\lambda(a, b) = b - a$ where $a < x < b$, more specifically $a = x - \frac{\varepsilon}{2}$ and $b = x + \frac{\varepsilon}{2}$. Now, $\lambda(a, b) = x + \frac{\varepsilon}{2} - \left(x - \frac{\varepsilon}{2}\right) = \varepsilon$ Since the singleton $\varepsilon > 0$ was arbitrary, taking the infimum, we get $\lambda^*(\{x\}) = 0$.

If $E \subset \mathbb{R}$ is countable, then we may write E as a countable union of singletons $E = \bigcup_{j=1}^{\infty} \{x_j\}$ and apply the countable subadditivity property to conclude that $m^*(E) = 0$. $\square$

Another intriguing example of a set with measure zero is the Cantor set. Mathematicians often joke that 97.3% of all counterexamples in real analysis involve the Cantor set. The cantor set denoted C, is constructed as the intersection of a sequence of sets C_n, where each set C_n is formed by iteratively removing middle thirds from the previous set, this process could be seen in Figure 1.6. The number of intervals in C_n is 2^n, and the length of each interval is $\ell(I) = \left(\frac{1}{3}\right)^n$. As the construction progresses, the intervals get smaller, and the Cantor set is defined as the intersection of all these sets $C = \cap_{n=1}^{\infty} C_n$, which results in the set of points that remain after infinitely many steps of removal.

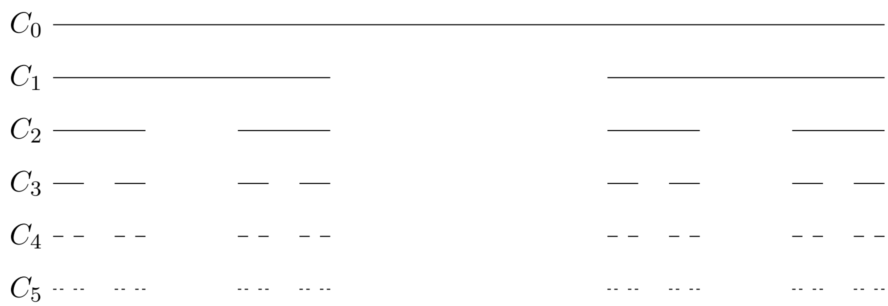

Figure 1.6: The Cantor set $C = \cap_{n=1}^{\infty} C_n$

We can immediately see that the Cantor set is non-empty as 0 and $1 \in C$, in fact, it is uncountable[4]. Now we will show that the Cantor set has zero Lebesgue measure.

Proof. On each step we removed 2^{k-1} intervals of length $\frac{1}{3}^k$ so

$$\mu(C) = \mu([0,1)) - (\frac{1}{2} + \frac{2}{9} + \dots) = \mu([0,1)) - \sum_{k=1}^{\infty} \frac{2^{k-1}}{3^k}$$

$$= 1 - \frac{1}{3} \sum_{k=0}^{\infty} \frac{2^k}{3^k} = 1 - \underbrace{\frac{1}{3} \frac{1}{1-\frac{2}{3}}}_{1} = 0 \ .$$

We conclude that the Cantor set has Lebesgue measure zero. □

We have seen a couple of examples of sets that are Lebesgue measurable and have measure zero, as they are composed of singletons. Since we cannot create an open interval with just one point, we have shown that $\mathcal{B} \subset M_\lambda$. In other words, the Borel σ-field is contained within the space of all Lebesgue measurable sets.

[4]Every element $x \in C$ can be expressed as $x = \sum_{k=1}^{\infty} \frac{a_k}{3^k}$, where $a_k \in \{0, 2\}$. The proof that the Cantor set is uncountable relies on the argument involving ternary expansions, which can be found online.

2 Functions, Random Variables, and Integration

In this chapter, we revisit familiar concepts such as functions, random variables, integration, and probability density functions within the measure-theoretic framework. This powerful approach allows us to uniformly apply definitions and theorems to both discrete and continuous random variables. Furthermore, measure theory significantly broadens our capacity for integration, enabling us to integrate functions that were previously inaccessible with traditional Riemann integrals. Additionally, we will explore what some consider to be most important results in analysis, notably the Dominated Convergence Theorem and the Fubini-Tonelli Theorem, emphasizing their crucial role and practical applications.

2.1 Simple Functions and Random Variables

We briefly introduce the concepts of a simple function and a simple random variable, laying the groundwork for measurable functions and random variables in general. Random variables give us a numerical description of the outcome and are typically denoted by capital Roman letters $(X, Y, Z \dots)$.

As usual, we begin with the definition. We will introduce *the indicator function*, which determines whether an element belongs to a specific set. This function is an important tool that we will use throughout this chapter.

Definition 2.1.1 (Indicator Function)**.** *Given $E \subseteq \mathbb{R}$, we let $\chi_E : \mathbb{R} \to \mathbb{R}$ denote the characteristic function of E, given by*

$$\chi_E(x) := \begin{cases} 1 & \text{if } x \in E, \\ 0 & \text{if } x \notin E. \end{cases}$$

Now we can use this idea to define *simple functions.* A function is called simple if it takes on only finitely many values.

Definition 2.1.2 (Simple Function)**.** *Suppose $(\mathbb{X}, \mathcal{F})$ is a measurable space. A function $f : \mathbb{X} \to \mathbb{R}$ is called a simple function if $c_1, \dots, c_n$ are distinct nonzero values and f has form*

$$f = c_1\chi_{E_1} + \cdots + c_n\chi_{E_n},$$

for pairwise disjoint $E_1, \dots, E_n \in \mathcal{F}$*.*

If such sets E_i are intervals $I_1, \dots, I_n \subseteq \mathbb{R}$, we and equivalently write $f(x) = \sum_{i=1}^{n} c_i \chi_{I_i}(x)$ for all $x \in \mathbb{R}$. Then it is easy to find integral of such simple functions as $\lambda\big(\chi_{I_j}(x)\big) = \lambda(I_j)$

Definition 2.1.3 (Integral of a non-negative simple function)**.** *Let f be a simple function such that $f = \sum_{j=1}^{k} c_j \chi_{I_j}(x)$ where $c_1, \dots, c_k \geq 0$, and $I_1, \dots, I_k \subseteq \mathbb{R}$ are pairwise disjoint intervals. Then, we define the integral of f as*

$$\int_{\mathbb{R}} \sum_{j=1}^{k} c_j \chi_{I_j}(x)\, dx := \sum_{j=1}^{k} c_j \lambda(I_j).$$

A *random variable*, simply put, is a mapping X that takes an element ω from the set Ω and maps it to its numerical representation x_i. For each $X(\omega) = x_i$, the corresponding ω must be an element of the σ-field, because we need to be able to work with such ω's.

The concept of a *simple random variable* is similar to that of a *discrete random variable*, with the key difference being that a simple random variable takes only a finite number of distinct values, while a discrete random variable is a broader category that includes all random variables with a countable set of possible values, whether finite or infinite.

Definition 2.1.4 (Simple Random Variable)**.** *Let $(\Omega, \mathcal{F}, P)$ be a probability space. Then X is a simple random variable if $X : \Omega \to \mathbb{R}$ is a real-valued function that only takes on a finite number of values $x_1, \dots, x_p$ and*

$$\{\omega \in \Omega : X(\omega) = x_i\} \in \mathcal{F}.$$

Whether or not X satisfies this condition depends only on X, not on P, but the point of the definition is to ensure that the probabilities $P(\omega : X(\omega) = x)$ are well-defined. One way to write such a function is to finitely partition Ω into disjoint sets $\{A_i\}_{i=1}^{p}$. That is,

$$\bigcup_{i=1}^{p} A_i = \Omega \quad \text{and} \quad A_i \cap A_j = \emptyset \quad \text{for } i \neq j,$$

and write

$$X(\omega) = \sum_{i=1}^{p} x_i \chi[\omega \in A_i].$$

Then the probability that $X = x_i$ can be equivalently written as

$$P(X = x_i) = P(\{\omega : X(\omega) = x_i\}) = P(A_i).$$

Furthermore, this allows us to define the expectation of the simple random variable X to be

$$\mathbb{E}[X] = \sum_{i=1}^{p} x_i P(X = x_i).$$

We will connect all of this in the following example.

Example 2.1.1 (Binary Steps). Let $\Omega = (0, 1]$ and $A_1 = (0, 0.25]$, $A_2 = (0.25, 0.5]$, $A_3 = (0.5, 0.75]$, $A_4 = (0.75, 1]$, and $x_i = \frac{i-1}{4}$. Then, for the probability space $((0, 1], \mathcal{B}, \lambda)$, $\lambda(A_i) = 0.25$ for any $i = 1, 2, 3, 4$. Thus, the simple random variable X as above takes on the values 0, 0.25, 0.5, 0.75 each with probability 25%. The expectation is

$$\mathbb{E}[X(\Omega)] = \frac{0.25 + 0.5 + 0.75}{4} = 0.375.$$

2.2 Measurable Functions and Random Variables

To extend the above idea of a simple random variable, we want to replace the finite x_i with any $x \in \mathbb{R}$ on the Borel set $\mathcal{B} \subset \mathbb{R}$. We will also consider general functions mapping from one measure space to another.

We begin with two measurable spaces $(\mathbb{X}, \mathcal{X})$ and $(\mathbb{Y}, \mathcal{Y})$. Let f be a function that maps from $\mathbb{X}$ to $\mathbb{Y}$, then we can consider f applied to sets. For $A \subset \mathbb{X}$ and $B \subset \mathbb{Y}$, we have

$$f(A) = \{y \in \mathbb{Y} : y = f(x) \text{ for some } x \in A\}$$
$$f^{-1}(B) = \{x \in \mathbb{X} : f(x) \in B \text{ for some } y \in B\}.$$

This allows us to define what it means to be a *measurable function.* We say that a function f is measurable if for any element in $\mathbb{Y}$, we can find its preimage and that preimage is inside our σ-field on $\mathbb{X}$. Typically, the σ-fields of interest are the Borel σ-fields, and it is sometimes written $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$. Moreover, the space $(Y, \mathcal{Y})$ is typically taken to be $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ or $(\mathbb{R}^+, \mathcal{B}(\mathbb{R}^+))$. In this case, we say that f is Borel measurable. If we replace $\mathcal{B}(\mathbb{R})$ with $M_\lambda(\mathbb{R})$, the set of Lebesgue measurable subsets of $\mathbb{R}$, then we say f is Lebesgue measurable. The set of Lebesgue measurable functions gives us a much larger collection of functions to define integrals on.

Definition 2.2.1 (Measurable Function)**.** *A function $f : \mathbb{X} \to \mathbb{Y}$ is said to be $\mathcal{X}-$measurable (or just measurable if the sigma field used is clear from the context) if*

$$f^{-1}(B) \in \mathcal{X} \quad \textit{for any } B \in \mathcal{Y}.$$

A random variable X is a measurable function, meaning X maps outcomes from the sample space $\mathbb{X}$ (equipped with a σ-algebra $\mathcal{X}$) to values in a measurable space $\mathbb{Y}$ (equipped with a σ-algebra $\mathcal{Y}$). This ensures that the preimages of measurable sets under X are measurable, i.e. for any $A \in \mathcal{Y}$, we have $X^{-1}(A) \in \mathcal{X}$, which is crucial for defining events and probabilities in the context of random variables (see figure ??).

This definition is powerful because it does not distinguish between discrete and continuous distributions. Instead, it establishes a general framework, and later we will demonstrate how to combine any two σ-finite measure spaces, and by extension, any finite number of σ-finite measure spaces.

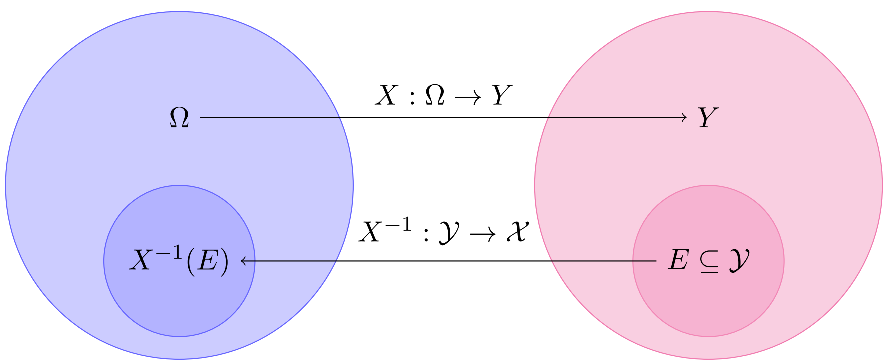

Figure 2.1: Pictorial representation of random variable

Definition 2.2.2 (Random Variable)**.** *Let $(\Omega, \mathcal{F}, P)$ be a probability space. A measurable function X on $(\Omega, \mathcal{F})$ is random variable if $X : \Omega \to \mathbb{R}$ and*

$$\{\omega \in \Omega : X(\omega) = x\} \in \mathcal{F}.$$

2.3 Integration

We have already seen that simple functions can be integrated. Now, we want to extend this idea to any measurable function. We will consider measurable functions mapping from $(\Omega, \mathcal{F})$ to $[-\infty, \infty]$, which is called the extended real line. This allows us to handle sets such as $f^{-1}(\infty)$, for example.

The integral of a non-negative measurable function, or *the Lebesgue integral*, is defined as the supremum taken over all non-negative simple functions. This is sufficiently strong, as for any general measurable function f, we can decompose it as $f = f^+ - f^-$, where f^+ and f^- are non-negative measurable functions.

Definition 2.3.1 (Integral of a Non-negative Measurable Function)**.** *Let $f : \mathbb{R} \to [0, \infty)$ be a non-negative measurable function, then its integral is*

$$\int_{\mathbb{R}} f(x)\, dx := \sup \left\{ \int_{\mathbb{R}} \sum_{j=1}^{k} c_j \chi_{I_j}(x)\, dx : \, 0 \leq \sum_{j=1}^{k} c_j \chi_{I_j}(x) \leq f \right\}.$$

Remark 2.3.1 (Dealing with Infinities)**.** In the above definition, we may need the following conventions:

$$0 \times \infty = 0, \quad c \times \infty = \infty \quad \text{for} \quad c > 0.$$

Thus, if $f = 0$ on a set A where $\mu(A) = \infty$, then that term in the above definition is just 0, as desired. Also, $\infty - \infty$ is undefined. Hence, $\int f\, d\mu$ is undefined if $\int f^+\, d\mu$ and $\int f^-\, d\mu$ are both ∞. In the case that both $\int f^+\, d\mu$ and $\int f^-\, d\mu$ are finite, we say that f is *integrable.*

Another strength of the measure-theoretic approach to integration is that, for our theorems, functions do not need to be exactly equal; they only need to be equal almost everywhere. This is because the integral does not account for differences on sets with measure zero. We start with a definition hence, we demonstrate this important property of integrals.

Definition 2.3.2 (Almost Everywhere / Almost Surely)**.** *Let $(\mathbb{X}, \mathcal{F}, \mu)$ be a measure space. For two functions $f, g : \mathbb{X} \to \mathbb{R}$, we say that $f = g$ a.e. (almost everywhere) when the set $N = \{\omega : f(\omega) \neq g(\omega)\}$ has measure $\mu(N) = 0$. In probability theory, "almost everywhere" is replaced with "almost surely," abbreviated a.s., and it is equivalently written "with probability 1" or wp1.*

This result essentially tells us that we can modify functions on a set of measure zero without affecting their integrability or the value of their integrals.

Theorem 2.3.2 (Integral of Functions Equal Almost Everywhere)**.** *Let* $(\mathbb{X}, \mathcal{F}, \mu)$ *be a measure space, and let* $f, g : \mathbb{X} \to [-\infty, \infty]$ *be measurable functions such that* $f = g$ *almost everywhere (a.e.). Then,* f *is integrable if and only if* g *is integrable. Furthermore, if both* f *and* g *are integrable, then*

$$\int f \, d\mu = \int g \, d\mu.$$

Proof. Let $f = g$ on $\mathbb{X} \backslash N$, where $\mu(N) = 0$. For any measurable function $h : \mathbb{X} \to [-\infty, \infty]$, we can consider when $\int_{\mathbb{X}} h \, d\mu$ and $\int_{\mathbb{X} \backslash N} h \, d\mu$ coincide. This is true when h is an indicator function, say, $h = 1_A$, because $\mu(A) = \mu(A \setminus N)$. Thus, the integrals are also equal when h is a simple function. We can extend this to non-negative simple functions h and hence to any non-negative measurable function. Finally, using $\int h \, d\mu = \int h^+ \, d\mu - \int h^- \, d\mu$, we can show that the integrals will coincide for any integrable measurable function. To complete the proof, we note that

$$\int f \, d\mu = \int_{\mathbb{X} \backslash N} f \, d\mu = \int_{\mathbb{X} \backslash N} g \, d\mu = \int g \, d\mu.$$

2.4 Three Important Convergence Theorems

In what follows, we are interested in how to handle a sequence of integrals $\int f_i \, d\mu$ of measurable functions as $i \to \infty$. Under what conditions does it converge to some $\int f \, d\mu$? From a probability perspective, $\int X_i \, d\mu$ is the expectation of some random variable X_i, so you can think of the following as theorems about convergence of the mean of a sequence of random variables.

The Monotone Convergence Theorem (MCT) is a fundamental result in measure theory that provides conditions under which the limit of a sequence of integrals can be interchanged with the integral of the limit function.

Theorem 2.4.1 (Monotone Convergence Theorem)**.** *Suppose* $(\mathbb{X}, \mathcal{F}, \mu)$ *is a measure space and* $0 \leq f_1 \leq f_2 \leq \cdots$ *is an increasing sequence of* $\mathcal{F}$*-measurable functions. Define* $f : \mathbb{X} \to [0, \infty]$ *by*

$$f(x) = \lim_{k \to \infty} f_k(x).$$

Then,

$$\lim_{k \to \infty} \int f_k \, d\mu = \int f \, d\mu.$$

Proof. First, we need to check that f is measurable. For $c \in \mathbb{R}$, we consider the sets $(c, \infty]$, which generate the Borel σ-field. Since $0 \leq f_1 \leq f_2 \leq \cdots$, we have $f^{-1}((c,\infty]) = \bigcup_{i=1}^{\infty} f_i^{-1}((c,\infty])$, and since $f_i^{-1}((c,\infty]) \in \mathcal{F}$, it follows that f is measurable.

Because $f_k(x) \leq f(x)$ for every $x \in \mathbb{X}$, we have $\int_{\mathbb{X}} f_k \, d\mu \leq \int_{\mathbb{X}} f \, d\mu$ for each $k \in \mathbb{Z}_+$ (by Lemma **??**). Thus, $\lim_{k\to\infty} \int_{\mathbb{X}} f_k \, d\mu \leq \int_{\mathbb{X}} f \, d\mu$.

To prove the inequality in the other direction, let h be a function such that $0 \leq h(x) \leq f(x)$ and let $\varepsilon > 0$. Now, we define $\mathbb{X}_n \subset \mathbb{X}$ as the set on which our monotonically increasing function f_n is greater than or equal to the scaled simple function $(1-\varepsilon)h(x)$, or in other terms,

$$\mathbb{X}_n = \{x \in \mathbb{X} \mid f_n(x) \geq (1-\varepsilon)h(x)\}.$$

Then $\mathbb{X}_1 \subseteq \mathbb{X}_2 \subseteq \cdots$ is an increasing sequence of sets in $\mathcal{F}$ whose union equals $\mathbb{X}$.

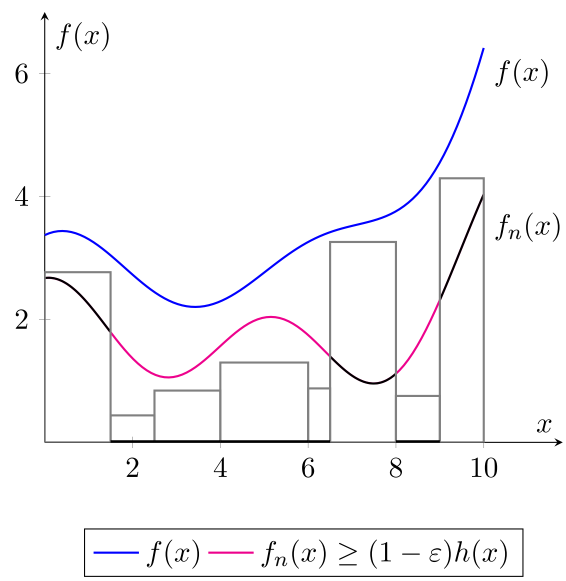

Figure 2.2: Your caption describing the figure goes here.

One could visualise such idea in (Figure 2.2) where boxes represent some simple function $(1-\varepsilon)h(x)$, the blue function is $f(x)$, under which is function $f_n(x)$ with its highlited parts that are above $(1-\varepsilon)h(x)$ on set $\mathbb{X}_n$ in which we are interested.

Then, the area under f_n on the whole $\mathbb{X}$ is greater than or equal to the are under f_n on $\mathbb{X}_n$ which is by construction greater than the are under $(1-\varepsilon)h(x)$ on $\mathbb{X}_n$

$$\int_{\mathbb{X}} f_n \, d\mu \geq \int_{\mathbb{X}_n} f_n \, d\mu \geq \int_{\mathbb{X}_n} (1-\varepsilon)h(x) \, d\mu.$$

Taking the limit as $n \to \infty$.

$$\lim_{n\to\infty} \int_{\mathbb{X}} f_n \, d\mu \geq \lim_{n\to\infty} \int_{\mathbb{X}_n} (1-\varepsilon)h(x) \, d\mu = \lim_{n\to\infty} \int_{\mathbb{X}} (1-\varepsilon)h(x) \, d\mu.$$

Since $\varepsilon > 0$ was arbitrary, we can take ε as small as we like, and therefore:

$$\int_{\mathbb{X}} f_n \, d\mu \geq \int_{\mathbb{X}_n} h(x) \, d\mu$$

Now, we can take the supremum of all such simple functions $h(x)$ and by (definition 2.3.1) we have that $\sup(h(x)) = f(x)$. Hence,

$$\int_{\mathbb{X}} f_n \, d\mu \geq \int_{\mathbb{X}_n} f(x) \, d\mu,$$

and result follows as desired. □

The Monotone Convergence Theorem (MCT) is particularly useful when dealing with sequences of measurable functions and their integrals. The general idea is that under certain conditions, the limit of integrals can be exchanged with the integral of the limit.

Example 2.4.2 (Application of MCT)**.** Let $(g_n)_{n\in\mathbb{N}}$ be a sequence of non-negative measurable functions with $g_n \to [0,\infty]$ for all $n \in \mathbb{N}$. Then,

$$\sum_{n=1}^{\infty} g_n \text{ on } \mathbb{X} \to [0,\infty] \text{ is also well defined measurable function.}$$

We also have the equality:

$$\int_X \sum_{n=1}^{\infty} g_n \, d\mu = \sum_{n=1}^{\infty} \int_X g_n \, d\mu.$$

For the next two convergence theorems we need a following definition.

Definition 2.4.1 (Limit Inferior and Limit Superior)**.** *The limit inferior of a sequence* (x_n) *is defined by*

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty}\left(\inf_{m\geq n} x_m\right) = \sup\left\{\inf\left\{x_m : m \geq n\right\} : n \geq 0\right\}.$$

Similarly,

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty}\left(\sup_{m\geq n} x_m\right) = \inf\left\{\sup\left\{x_m : m \geq n\right\} : n \geq 0\right\}.$$

When looking at Figure 2.3, we can notice that the limit inferior is increasing as $n \to \infty$, that is, one could imagine "cutting off" terms from the left and then finding a new inferior value for our sequence. Similarly for the limit superior.

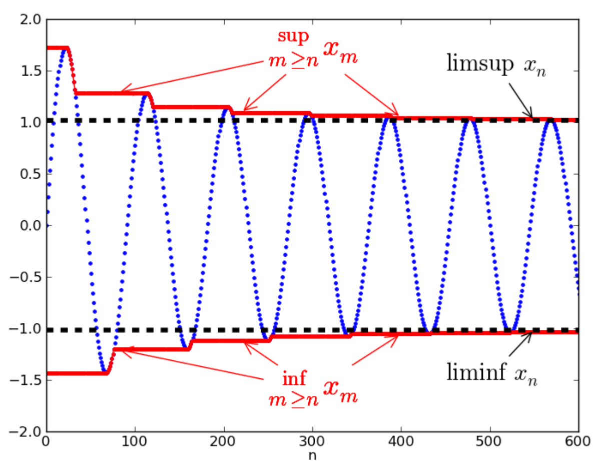

Figure 2.3: Limit inferior and limit superior

The next convergence theorem, Fatou's Lemma, provides an inequality that relates the integral of the pointwise limit of a sequence of functions to the limit inferior of the integrals of the functions in the sequence. Specifically, it tells us that the integral of the pointwise limit is greater than or equal to the limit inferior of the integrals. Fatou's Lemma can be applied even when the sequence of functions is not increasing, which generalizes the idea of exchanging limits and integrals. This makes it particularly useful for sequences that do not satisfy the conditions required by the Monotone Convergence Theorem (MCT). In probability theory, Fatou's Lemma plays a crucial role in handling the expectation of the pointwise limit of random variables. This has important implications for results in probability, such as the law of large numbers and the convergence of expectations.

Theorem 2.4.3 (Fatou's Lemma)**.** *Let* $(\mathbb{X}, \mathcal{F}, \mu)$ *be a measure space and let* $\{f_i\}_{i=1}^{\infty}$ *be non-negative measurable functions from* $\mathbb{X}$ *to* $\mathbb{R}$*. Then,*

$$\int \liminf_{i\to\infty} f_i \, d\mu \leq \liminf_{i\to\infty} \int f_i \, d\mu.$$

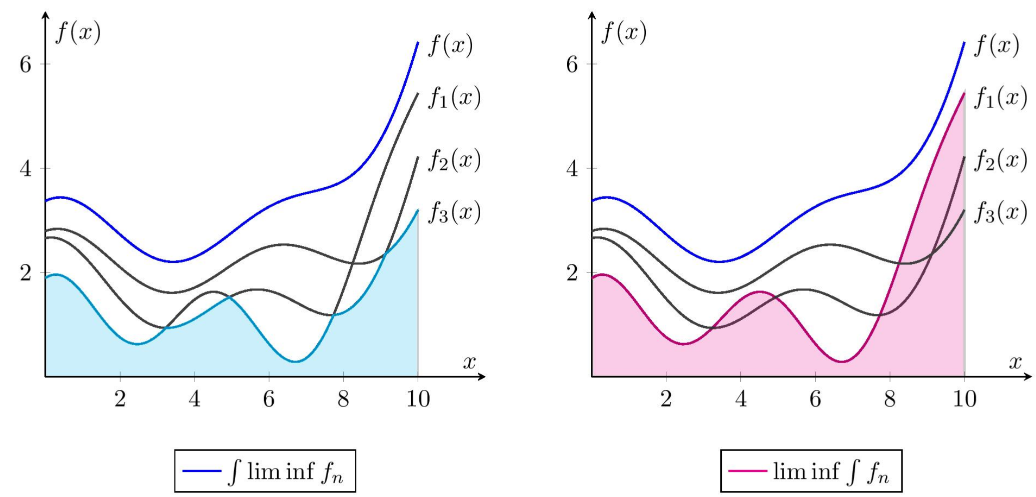

Figure 2.4: Comparing $\int \liminf f_n$ and $\liminf \inf f_n$.

Such an idea could be captured in Figure 2.4. The left picture illustrates an example of $\int \liminf f_n$, where we examine which f_n for $n > m$ has the smallest value in a pointwise manner. On the other hand, the right picture represents $\liminf \int f_n$, where one can imagine the integrals as a sequence of real numbers, and we are taking the smallest value.

Proof. We will define

$$g := \liminf_{n\to\infty} x_n := \lim_{n\to\infty} \overbrace{\left(\inf_{m\geq n} x_m\right)}^{g_n}.$$

Then $0 \leq g_1 \leq g_2 \leq \cdots$ is increasing sequence of $\mathcal{F}$-measurable functions.

Using Monotone Convergence Theorem and definition of g we get following equalities.

$$\int \lim_{n\to\infty} g_n \, d\mu \overset{\text{by MCT}}{=} \lim_{n\to\infty} \int g_n \, d\mu = \liminf_{n\to\infty} \int g_n \, d\mu$$

We know that $g_n \leq f_n$ for all $n \in \mathbb{N}$. Then by Lemma **??** we have $\int_X g_n \, d\mu \leq \int_X f_n \, d\mu$. Then,

$$\int \lim_{n\to\infty} g_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu. \tag{2.1}$$

Substituting f_n back into the inequality (2.1)

$$\int \liminf_{n\to\infty} f_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.$$

By combining the inequalities, the result follows. □

Fatou's Lemma serves as a building block for the *Dominated Convergence Theorem (DCT)*, which is a more refined result providing conditions under which the limit and integral can be interchanged. Similar to Fatou's Lemma, the Dominated Convergence Theorem does not require the sequence of functions to be monotonic. Instead, it requires the existence of an integrable function g such that $|f_n| \leq g$ for all n. Under this condition, the limit of the integrals equals the integral of the limit function.

Theorem 2.4.4 (Dominated Convergence)**.** *Suppose $(\mathbb{X}, \mathcal{F}, \mu)$ is a measure space, $f : \mathbb{X} \to [-\infty, \infty]$ is $\mathcal{F}$-measurable, and $f_1, f_2, \ldots$ are $\mathcal{F}$-measurable functions from $\mathbb{X}$ to $[-\infty, \infty]$ such that $\lim_{n\to\infty} f_n(x) = f(x)$ for almost every $x \in \mathbb{X}$. If there exists an $\mathcal{F}$-measurable function $g : \mathbb{X} \to [0, \infty]$ such that $\int g \, d\mu < \infty$ and $|f_n(x)| \leq g(x)$ almost every $x \in \mathbb{X}$ for every $n \in \mathbb{N}$, then*

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

Proof. First we will show that $|f|$ is a $\mathcal{F}$-measurable function.

$$\text{If } \ |f_n| \leq g \ \ \forall \ n \ \ \text{a.e.} \ \implies \ |f| \leq g. \tag{2.2}$$

We dont know much about sequence $\{f_n\}_{n\in\mathbb{N}}$ besides that it eventually converges to f. If $\{f_n\}_{n\in\mathbb{N}}$ was monotonic, we could apply MCT. Since we do not know that such conditions hold, we will try and rearrange 2.2 to get the following inequalities.

$$\implies -g \leq f \leq g$$

$$\implies -g - f \leq 0 \leq g - f.$$

Inequality I: $\quad g + f \geq 0.$ Inequality II: $\quad g - f \geq 0.$

Now, inequalities I & II we have non-negative measurable functions so we can apply Fatou's Lemma. Now,

$$\begin{aligned}\int g\,d\mu + \int f\,d\mu &= \int (g+f)\,d\mu \\ &\overset{\text{by Fatou}}{\leq} \liminf_{n\to\infty} \int (g+f_n)\,d\mu \\ &= \int g\,d\mu + \liminf_{n\to\infty} \int f_n\,d\mu.\end{aligned}$$

$$\begin{aligned}\int g\,d\mu - \int f\,d\mu &= \int (g-f)\,d\mu \\ &\overset{\text{by Fatou}}{\leq} \liminf_{n\to\infty} \int (g-f_n)\,d\mu \\ &= \int g\,d\mu - \limsup_{n\to\infty} \int f_n\,d\mu.\end{aligned}$$

By canceling $\int g\ d\mu$ from inequalities I & II we get

$$\int f\,d\mu \leq \liminf_{n\to\infty} \int f_n\,d\mu \qquad \text{and} \qquad \limsup_{n\to\infty} \int f_n\,d\mu \leq \int f\,d\mu.$$

Then,

$$\limsup_{n\to\infty} \int f_n\,d\mu \leq \int f\,d\mu \leq \liminf_{n\to\infty} \int f_n\,d\mu.$$

We have shown that the limit exist and

$$\int f\,d\mu = \lim_{n\to\infty} \int f_n\,d\mu.$$

Result follows as desired. □

We proved Monotone Convergence Theorem (MCT), Fatou's Lemma, and the Dominated Convergence Theorem (DCT). These theorems are crucial for handling limits under integration, especially in contexts involving sequences of functions. The MCT ensures that when dealing with increasing, non-negative sequences of functions, the limit can be interchanged with the integral. Fatou's Lemma provides a general inequality that connects the limit inferior of a sequence of functions to the integral, offering a more flexible tool in situations where pointwise convergence may not be strict. The DCT, one of the most powerful tools in integration theory, allows the interchange of the limit and integral when functions are dominated by an integrable function, extending the scope of MCT to broader settings.

These results are essential for establishing robust frameworks in probability and analysis, providing key tools for dealing with limits in measure-theoretic contexts.

2.5 Pushforward Measures and Their Role in Probability

In probability theory, we often want to shift attention from the sample space to the real line, where our random variable lives. This section formalizes how a measurable function transforms a measure — a key idea in expressing probabilities and expectations directly in terms of distributions.

When working with two measurable spaces $(\mathbb{X}, \mathcal{A})$ and $(\mathbb{Y}, \mathcal{C})$, a measure μ on the first space, together with a measurable function $h : \mathbb{X} \to \mathbb{Y}$, allows us to construct a measure on the second space. This is done by defining the *image measure* $\nu = \mu \circ h^{-1}$, as illustrated in Figure 2.5.

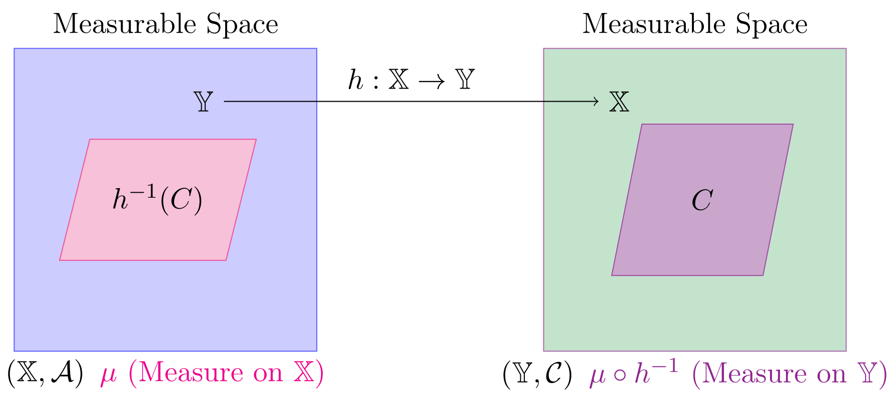

Figure 2.5: Pictorial representation image measure

Definition 2.5.1 (Image Measure). *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f : X \to Y$ be a measurable function into a measurable space $(Y, \mathcal{B})$. Define a measure $\nu : \mathcal{B} \to [0, \infty]$ on $(Y, \mathcal{B})$ by*

$$\nu(B) := \mu(f^{-1}(B)) \quad \textit{for all } B \in \mathcal{B}.$$

Then $\nu = \mu \circ f^{-1}$ is called the image measure *of μ under f.*

In probability theory, the image measure associated with a random variable is known as its distribution, or law.

Definition 2.5.2 (Distribution of a Random Variable). *Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $X : \Omega \to \mathbb{R}$ be a random variable. The* distribution *(or* law*) of X, denoted μ_X, is the measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ defined by*

$$\mu_X(B) := P(X^{-1}(B)) \quad \text{for all } B \in \mathcal{B}(\mathbb{R}).$$

Equivalently, $\mu_X = P \circ X^{-1}$.

In probability theory, the change of variable theorem provides the formal justification for expressing expectations in terms of a random variable's distribution, rather than over the original probability space.

Theorem 2.5.1 (Change of Variable). *Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $X : \Omega \to \mathbb{R}$ be a measurable function. Let $\mu_X = P \circ X^{-1}$ be the distribution of X. If $h : \mathbb{R} \to \mathbb{R}$ is a measurable function such that $h \circ X$ is integrable, then*

$$\int_\Omega h(X)\, dP = \int_\mathbb{R} h(x)\, d\mu_X(x).$$

Proof. As usual, we begin with the set function case, then extend to simple functions, then to non-negative measurable functions, and finally to arbitrary measurable functions.

Step (1). Let χ_C be the indicator function of a measurable set $C \subseteq \mathbb{Y}$. Then:

$$\int_\mathbb{Y} \chi_C\, d(h_*\mu) = (h_*\mu)(C) = \mu(h^{-1}(C)),$$

and

$$\int_\mathbb{X} \chi_C \circ h\, d\mu = \int_\mathbb{X} \chi_C(h(x))\, d\mu(x) = \int_\mathbb{X} \chi_{h^{-1}(C)}(x)\, d\mu(x).$$

Step (2). Let g be a simple function, i.e., $g = \sum_{i=1}^n c_i \chi_{C_i}$. Then:

$$\int_\mathbb{Y} g\, d(h_*\mu) = \sum_{i=1}^n c_i \int_\mathbb{Y} \chi_{C_i}\, d(h_*\mu) \overset{\text{Step (1)}}{=} \sum_{i=1}^n c_i \int_\mathbb{X} \chi_{C_i}(h(x))\, d\mu(x).$$

$$= \int_\mathbb{X} \left(\sum_{i=1}^n c_i \chi_{C_i}(h(x)) \right) d\mu(x) = \int_\mathbb{X} g \circ h\, d\mu.$$

Step (3). Let $g : \mathbb{Y} \to [0, \infty)$ be a measurable function. Then, by the definition of the

Lebesgue integral:

$$\begin{aligned}
\int_{\mathbb{Y}} g\, d(h_*\mu) &= \sup\left\{\int_{\mathbb{Y}} \tilde{s}\, d(h_*\mu) : 0 \le \tilde{s} \le f\right\} \\
&= \sup\left\{\int_{\mathbb{X}} \tilde{s}\circ h\, d\mu : 0 \le \tilde{s}\circ h \le f\right\} \\
&= \sup\left\{\int_{\mathbb{X}} s\, d\mu : 0 \le s \le f\right\} \\
&= \int_{\mathbb{X}} g\circ h\, d\mu.
\end{aligned}$$

Step (4). For a general measurable function $g : \mathbb{Y} \to \mathbb{R}$, write $g = g^+ - g^-$, where both g^+ and g^- are non-negative measurable functions. By linearity and the result for non-negative functions, we conclude that the change of variable formula holds for all measurable functions. □

The expectation $\mathbb{E}[X]$ of a random variable X is the average or central value of X.

Definition 2.5.3 (Expectation)**.** *Let* $(\Omega, \mathcal{F}, P)$ *be a probability space, and let* $X : \Omega \to \mathbb{R}$ *be an integrable random variable. The* expectation *of* X *is defined as*

$$\mathbb{E}[X] := \int_{\Omega} X\, dP.$$

Since the distribution captures how values of a random variable are spread over $\mathbb{R}$, it is natural to ask whether quantities like expectation can be expressed in terms of this distribution. The change of variable theorem provides the answer.

Proposition 2.5.1 (Expectation with Respect to the Law of a Random Variable)**.** *Let* $X : \Omega \to \mathbb{R}$ *be an integrable random variable with distribution* $\mu_X = P \circ X^{-1}$*. Then*

$$\mathbb{E}[X] = \int_{\mathbb{R}} x\, d\mu_X(x).$$

To understand when one measure can be expressed in terms of another, we first introduce two fundamental relationships between measures: absolute continuity and singularity. Absolute continuity is the key condition in the Radon–Nikodym theorem, which ensures that a measure can be written as an integral against a measurable function — the Radon–Nikodym derivative. In contrast, singularity describes the case where no such representation is possible, as the measures are supported on disjoint sets.

Definition 2.5.4 (Absolutely Continuous and Singular Measures)**.** *Let $(X, \mathcal{F})$ be a measurable space, and let μ and λ be two measures on $\mathcal{F}$.*

- We say that μ is absolutely continuous with respect to λ, denoted $\mu \ll \lambda$, if for every measurable set $A \in \mathcal{F}$,

$$\lambda(A) = 0 \quad \textit{implies} \quad \mu(A) = 0.$$

- We say that μ is singular with respect to λ, denoted $\mu \perp \lambda$, if there exists a measurable set $B \in \mathcal{F}$ such that:

$$\lambda(B) = 0 \quad \textit{and} \quad \mu(B^c) = 0.$$

This means that μ and λ are concentrated on disjoint measurable sets.

Example 2.5.2 (Absolutely Continuous Measure)**.** Consider the *zero measure* (or *null measure*), which we denote by $\mathbf{0}$. This measure assigns zero to every measurable set. That is, for any measurable space $(\Omega, \mathcal{F})$, it satisfies

$$\mathbf{0}(A) = 0 \quad \text{for all } A \in \mathcal{F}.$$

Since the zero measure is always zero, it is absolutely continuous with respect to any measure μ. Indeed, if $\mu(A) = 0$, then clearly $\mathbf{0}(A) = 0$ as well, meaning that $\mathbf{0} \ll \mu$.

Example 2.5.3 (Singular Measure)**.** Consider the Dirac measure at 0, denoted δ_0, which is defined by:

$$\delta_0(A) = \begin{cases} 1, & \text{if } 0 \in A, \\ 0, & \text{if } 0 \notin A. \end{cases}$$

The Dirac measure δ_0 is singular with respect to the Lebesgue measure λ on $\mathbb{R}$. Specifically, we have:

$$\lambda(\{0\}) = 0, \quad \text{and} \quad \delta_0(\mathbb{R} \setminus \{0\}) = 0.$$

Thus, δ_0 and λ are mutually singular, and we write $\delta_0 \perp \lambda$.

The concept of absolute continuity plays a central role in understanding when one measure can be represented in terms of another. The Radon–Nikodym theorem formalizes this relationship by stating that, under absolute continuity, a measure can be expressed as an integral with respect to another measure using a measurable function — the Radon–Nikodym derivative.

This theorem provides a powerful generalization of the idea of density functions in probability theory, allowing for a rigorous formulation of concepts such as probability densities, likelihoods, and change of measure techniques.

Theorem 2.5.4 (Radon-Nikodym Theorem)**.** *Let $(\Omega, \mathcal{F})$ be a measurable space, and let μ and ν be two σ-finite measures on $\mathcal{F}$. If ν is absolutely continuous with respect to μ then there exists a measurable function $f : \Omega \to [0, \infty)$ such that for all $A \in \mathcal{F}$,*

$$\nu(A) = \int_A f \, d\mu.$$

The function f is called the Radon-Nikodym derivative of ν with respect to μ and is denoted by

$$f = \frac{d\nu}{d\mu}.$$

When the distribution of a random variable is absolutely continuous with respect to Lebesgue measure, the Radon–Nikodym theorem guarantees the existence of a measurable function, called a probability density function (PDF), that fully characterizes the distribution. This allows us to recover the entire distribution from a function on $\mathbb{R}$.

Definition 2.5.5 (Probability Density Function)**.** *Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $X : \Omega \to \mathbb{R}$ be a random variable. The probability distribution of X is the pushforward measure μ_X on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, defined by*

$$\mu_X(B) = P(X \in B), \quad \text{for all } B \in \mathcal{B}(\mathbb{R}).$$

If μ_X is absolutely continuous with respect to the Lebesgue measure λ, that is, $\mu_X \ll \lambda$, then by the Radon-Nikodym theorem, there exists a Lebesgue-integrable function $f_X : \mathbb{R} \to [0, \infty)$ such that for all $B \in \mathcal{B}(\mathbb{R})$,

$$P(X \in B) = \mu_X(B) = \int_B f_X(x) \, d\lambda(x) = \int_B f_X(x) \, dx.$$

The function f_X is called the probability density function (PDF) of X.

2.6 Product Measures and the Fubini-Tonelli Theorem

Now that we have defined the integral, we can more formally construct product measures and integrals over product spaces. First, given two σ-fields $\mathcal{X}$ and $\mathcal{Y}$, we will denote the product σ-field to be $\mathcal{X} \times \mathcal{Y}$, which is the σ-field generated by the sets $A \times B$ for $A \in \mathcal{X}$ and $B \in \mathcal{Y}$. Sets of the form $A \times B$ for $A \in \mathcal{X}$ and $B \in \mathcal{Y}$ are called rectangles. The collection of all rectangles will be denoted as $\mathcal{R}$.

The Fubini-Tonelli Theorem provides a powerful result for integrating over product spaces. In particular, it justifies swapping the order of integration for the double integrals of functions defined on the product σ-fields, under certain conditions. This is crucial, because joint probabilities often require computing double integrals. One such application is found in the proof of the Central Limit Theorem. The Fubini-Tonelli Theorem follows closely from the existence and uniqueness of the product measure. So, our goal is to prove the following theorem:

Theorem 2.6.1 (Existence and Uniqueness of Product Measure)**.** *Let* $(\mathbb{X}, \mathcal{X}, \mu)$ *and* $(\mathbb{Y}, \mathcal{Y}, \nu)$ *be* σ*-finite measure spaces. Let* π *be a set function on* $\mathbb{X} \times \mathbb{Y}$ *such that for* $A \in \mathcal{X}$ *and* $B \in \mathcal{Y}$*, we have* $\pi(A \times B) = \mu(A)\nu(B)$*. Then,* π *extends uniquely to a measure on* $(\mathbb{X} \times \mathbb{Y}, \mathcal{X} \times \mathcal{Y})$ *such that for any* $E \in \mathcal{X} \times \mathcal{Y}$*,*

$$\pi(E) = \int \int \chi_E(x, y)\, d\mu(x)\, d\nu(y) = \int \int \chi_E(x, y)\, d\nu(y)\, d\mu(x).$$

We will approach the proof of this theorem by first proving the result for finite measures μ and ν and then extending it to σ-finite measures. We first define monotone classes and prove an important theorem that we will use in the process.

Definition 2.6.1 (Monotone Class)**.** *A collection of subsets* $\mathcal{M}$ *of* Ω *is said to be monotone if*

i for $\{A_i\}_{i=1}^{\infty}$ *such that* $A_i \in \mathcal{M}$ *and* $A_i \uparrow A = \bigcup_{i=1}^{\infty} A_i$*, then* $A \in \mathcal{M}$*,*

ii for $\{A_i\}_{i=1}^{\infty}$ *such that* $A_i \in \mathcal{M}$ *and* $A_i \downarrow A = \bigcap_{i=1}^{\infty} A_i$*, then* $A \in \mathcal{M}$*,*

where the notation $A_i \uparrow A$ *means that we have* $A_i \subseteq A_{i+1}$ *for all* i *and* $\bigcup_{i=1}^{\infty} A_i \to A$*, and similarly* $A_i \downarrow A$ *means that* $A_{i+1} \subseteq A_i$ *for all* i *and* $\bigcap_{i=1}^{\infty} A_i \to A$*.*[1]

Recall that we defined a field to be such that $\emptyset, \Omega \in \mathcal{A}$, if $B, A \in \mathcal{A}$ then $B \setminus A \in \mathcal{A}$, and if $B, A \in \mathcal{A}$ then $A \cup B \in \mathcal{A}$. Instead, we can replace $B \setminus A \in \mathcal{A}$ with $A^c \in \mathcal{A}$. This is because $B \setminus A = B \cap A^c$.

[1] Note that we will also use this notation for monotonic sequences of real numbers and monotonic sequences of functions, in both cases implying monotonic convergence.

Theorem 2.6.2 (Monotone Class Theorem)**.** [2] *Let $\mathcal{A}$ be a field and $\mathcal{M}$ be monotone such that $\mathcal{A} \subseteq \mathcal{M}$. Then, $\sigma(\mathcal{A}) \subseteq \mathcal{M}$.*

Monotone classes may seem restricted since they are σ-fields with a very specific structure. However, we can leverage them to move one step further in showing that the product measure exists and is unique on σ-finite measure spaces. We do this by proving a lemma that allows us to swap the above order of integration for finite measures.

Lemma 2.6.3 (Fubini on Measurable Rectangles)**.** *Let $(\mathbb{X}, \mathcal{X}, \mu)$ and $(\mathbb{Y}, \mathcal{Y}, \nu)$ be finite measure spaces, and let*

$$\mathcal{F} = \left\{ E \subset \mathbb{X} \times \mathbb{Y} : \iint \chi_E \, d\mu(x) \, d\nu(y) = \iint \chi_E \, d\nu(y) \, d\mu(x) \right\}.$$

Then, $\mathcal{X} \times \mathcal{Y} \subset \mathcal{F}$.

Proof. It can be shown that the set function π where $\pi(A \times B) = \mu(A)\nu(B)$ is countably additive on $\mathcal{R}$.[3] First, let $E = A \times B$ for $A \in \mathcal{X}$ and $B \in \mathcal{Y}$, i.e. $E \in \mathcal{R}$. Then,

$$\begin{aligned} \int \int \chi_E(x, y) \, d\mu(x) \, d\nu(y) &= \mu(A) \int \chi_B(y) \, d\nu(y) = \mu(A)\nu(B) \\ &= \nu(B)\mu(A) = \nu(B) \int \chi_A(x) \, d\mu(x) = \int \int \chi_E(x, y) \, d\nu(y) \, d\mu(x). \end{aligned}$$

Therefore, $\mathcal{R} \subset \mathcal{F}$. Also, for disjoint $R_1, R_2 \in \mathcal{R}$, $\chi_{R_1 \cup R_2} = \chi_{R_1} + \chi_{R_2}$. Hence, $\mathcal{F}$ contains finite disjoint unions of rectangles. This implies that the field generated by the set of rectangles $\mathcal{A}$ is a subset of $\mathcal{F}$.

We next consider $\{E_i\}_{i=1}^{\infty}$ with $E_i \in \mathcal{F}$. If $E_i \uparrow E$, then monotone convergence implies that

$$\int \int \chi_{E_i}(x, y) \, d\mu(x) \, d\nu(y) \uparrow \int \int \chi_E(x, y) \, d\mu(x) \, d\nu(y)$$

and

$$\int \int \chi_{E_i}(x, y) \, d\nu(y) \, d\mu(x) \uparrow \int \int \chi_E(x, y) \, d\nu(y) \, d\mu(x).$$

Thus, $E \in \mathcal{F}$, and the same holds if $E_i \downarrow E$. Therefore, $\mathcal{F}$ is a monotone class. Finally, applying the monotone class theorem shows that $\mathcal{X} \times \mathcal{Y} = \sigma(\mathcal{A}) \subset \mathcal{F}$. □

[2] Proof moved to Appendix 2 due to space limitation

[3] By including finite unions of rectangles, the collection $\mathcal{R}$ can be extended to a field $\mathcal{A}$.

We are finally ready to prove Existence and Uniqueness of Product Measure.

Proof of Theorem 2.6.1. We first consider the case that μ and ν are finite measures. We begin with $\pi(A \times B) = \mu(A)\nu(B)$ for $A \times B \in \mathcal{R}$. Then, we extend π to the set function

$$\pi(E) := \int \int \chi_E(x, y)\, d\mu(x)\, d\nu(y)$$

for any $E \in \mathcal{X} \times \mathcal{Y}$. The above lemma says that we can define this set function and the order of integration can be reversed for any $E \in \mathcal{X} \times \mathcal{Y}$. Linearity of the integral implies that the set function π is finitely additive, and monotone convergence further implies that π is countably additive. Thus, π is a measure on $\mathbb{X} \times \mathbb{Y}$.

To show that π is unique, let ρ be some other set function such that $\rho(A \times B) = \mu(A)\nu(B)$ for $A \times B \in \mathcal{R}$. Let $\mathcal{M} = \{E \subset \mathbb{X} \times \mathbb{Y} : \pi(E) = \rho(E)\}$. Then, $\mathcal{M}$ is a monotone class, because for $E_i \uparrow E = \bigcup_{i=1}^{\infty} E_i$, we can rewrite $E = \bigcup_{i=1}^{\infty} D_i$ where $D_1 = E_1$ and $D_i = E_i \setminus E_{i-1}$ for $i \geq 2$ are disjoint. Thus, by countable additivity, $\pi(E) = \rho(E)$, so $E \in \mathcal{M}$. Arguing similarly for $E_i \downarrow E$ shows that $\mathcal{M}$ is monotone. Hence, application of the monotone class theorem implies that $\mathcal{X} \times \mathcal{Y} \subset \mathcal{M}$. Thus, π is unique on $\mathbb{X} \times \mathbb{Y}$ for finite measures μ and ν.

Now let μ and ν be σ-finite measures. Let $\{A_i\}_{i=1}^{\infty}$ and $\{B_i\}_{i=1}^{\infty}$ be disjoint partitions of $\mathbb{X}$ and $\mathbb{Y}$, respectively, such that $\mu(A_i) < \infty$ and $\nu(B_i) < \infty$. Then, for any $E \in \mathcal{X} \times \mathcal{Y}$, we define $E_{ij} = E \cap (A_i \times B_j)$. We are slicing E into disjoint rectangle which could be imagined as Figure 2.6.

Recall that σ-finite measures do not require the spaces $\mathbb{X}$ and $\mathbb{Y}$ to be finite. Instead, they only require that $\mu(A_i) < \infty$ for all i and $\nu(B_j) < \infty$ for all j. Thus, from the above finite measure case,

$$\int \int \chi_{E_{i,j}}(x, y)\, d\mu(x)\, d\nu(y) = \int \int \chi_{E_{i,j}}(x, y)\, d\nu(y)\, d\mu(x).$$

Summing over all i and j, we can extend π using monotone convergence again to get

$$\pi(E) = \int \int \chi_E(x, y)\, d\mu(x)\, d\nu(y) = \int \int \chi_E(x, y)\, d\nu(y)\, d\mu(x)$$

for any $E \in \mathcal{X} \times \mathcal{Y}$. Monotone convergence implies that π is countably additive and hence a measure on $\mathbb{X} \times \mathbb{Y}$. For any other measure ρ such that $\rho(A \times B) = \mu(A)\nu(B)$, countable

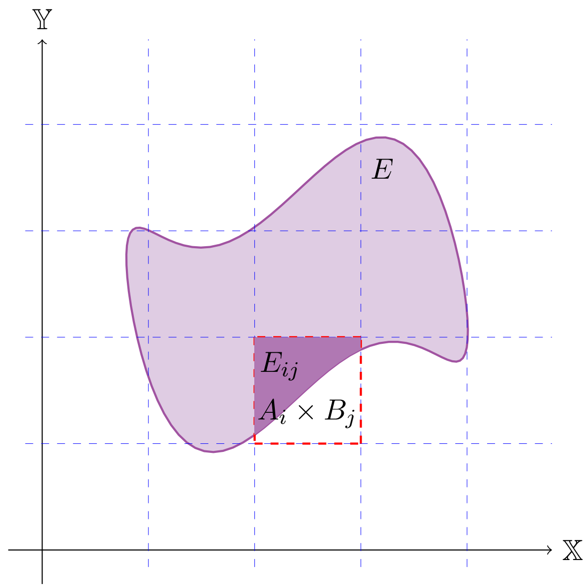

Figure 2.6: Pictorial representation of $E_{ij} = E \cap (A_i \times B_j)$

additivity and uniqueness for finite measures implies that

$$\pi(E) = \sum_{i,j} \pi(E_{i,j}) = \sum_{i,j} \rho(E_{i,j}) = \rho(E)$$

for any $E \in \mathcal{X} \times \mathcal{Y}$. Hence, the extension of π to $\mathbb{X} \times \mathbb{Y}$ is unique. $\square$

The existence and uniqueness of the product measure has a remarkable result. That is, it allows us to justify swapping the order of integration for many measurable functions defined on a σ-finite product space. Although very common and seemingly trivial in application, this is an important fact that we need to formally prove.

Theorem 2.6.4 (The Fubini-Tonelli Theorem)**.** *Let* $(\mathbb{X}, \mathcal{X}, \mu)$ *and* $(\mathbb{Y}, \mathcal{Y}, \nu)$ *be* σ*-finite measure spaces, and let* $f : \mathbb{X} \times \mathbb{Y} \to \mathbb{R}$ *be measurable with respect to* $\mathcal{X} \times \mathcal{Y}$ *such that either* $f \geq 0$ *or* $\iint |f| \, d(\mu \times \nu) < \infty$*. Then:*

$$\int f \, d(\mu \times \nu) = \int \int f(x, y) \, d\mu(x) \, d\nu(y) = \int \int f(x, y) \, d\nu(y) \, d\mu(x).$$

Also, $\int f(x, y) \, d\mu(x)$ *is* $\mathcal{Y}$*-measurable and* $\int f(x, y) \, d\nu(y)$ *is* $\mathcal{X}$*-measurable.*

Proof. This result is immediate if f is a simple function due to the above existence and uniqueness theorem for product measure and the linearity of the integral. That is,

$$\int\int_{\mathbb{X}_n} \chi_{E_i}(x,y)\,d\mu(x)\,d\nu(y) = \int\int_{\mathbb{X}_n} \chi_{E_i}(x,y)\,d\nu(y)\,d\mu(x).$$

Secondly, applying the monotone convergence theorem to simple functions implies that the above holds for non-negative measurable functions.

If instead we assume that $\int\int |f|\,d(\mu\times\nu) < \infty$, then $f = f^+ - f^-$ and the above holds for both f^+ and f^-. That is, both

$$\int f^+(x,y)\,d\mu(x) < \infty \text{ for } \nu\text{-almost-every } y \quad \text{and} \quad \int f^+(x,y)\,d\nu(y) < \infty \text{ for } \mu\text{-almost-every } x,$$

and similarly for f^-. Therefore,

$$\int |f|(x,y)\,d\nu(y) = \int f^+(x,y)\,d\nu(y) + \int f^-(x,y)\,d\nu(y) < \infty \quad (\mu \text{ a.e.}),$$

and thus

$$\int f(x,y)\,d\nu(y) = \int f^+(x,y)\,d\nu(y) - \int f^-(x,y)\,d\nu(y) < \infty \quad (\mu \text{ a.e.}).$$

As Theorem 2.3.3 tells us that we only require finiteness to occur almost everywhere to have the integral exist, we can integrate both sides of the above with respect to μ to get

$$\int\int f(x,y)\,d\nu(y)\,d\mu(x) = \int\int f^+(x,y)\,d\nu(y)\,d\mu(x) - \int\int f^-(x,y)\,d\nu(y)\,d\mu(x).$$

The same can be done swapping the role of μ and ν to conclude the theorem. □

3 Probability Theory

Having established the foundational tools of integration, convergence, and measure transformation, we now turn our attention to a cornerstone of probability theory: the Central Limit Theorem (CLT). Our objective is to employ the measure-theoretic framework developed thus far to rigorously formulate and prove a version of the CLT known as the *Lindeberg–Feller theorem*.

To build toward this goal, we will first examine how random variables combine, beginning with their joint behavior and the resulting distributions. This will naturally lead us to the concept of *convolution*, a fundamental operation that underpins the distribution of sums of independent random variables.

3.1 Joint Distributions and Convolution

In this section we seek to explore the joint behavior of two random variables with probability density functions f_x and f_y. To build intuition, we will first examine a simple case, rolling two fair dice and summing their outcomes, before progressing to more general settings.

Example 3.1.1 (Adding Two Fair Dice)**.** Let $\mathbb{X}_1 = \mathbb{X}_2 = \{1, 2, 3, 4, 5, 6\}$ and $P(x_i) = \frac{1}{6}$ for all $x_i \in \mathbb{X}_1 = \mathbb{X}_2$. Our random variables X_1 and X_2 each represent the outcomes of a 6-sided fair dice. Further, $\mathbb{X}_1 \times \mathbb{X}_2 = \{(x_1, x_2) \mid x_1, x_2 \in \{1, 2, 3, 4, 5, 6\}\}$ is our event space, with $P(x_j) = \frac{1}{36}$ for all $x_j \in \mathbb{X}_1 \times \mathbb{X}_2$. Now, the random variable $Z = X_1 + X_2$ represents the sum of the two dice. Then, $P(Z = z) = \frac{\text{Number of ways to achieve sum } z}{36} \quad \text{for} \quad z \in \{2, 3, \ldots, 12\}$.

All possible outcomes of experiment 3.1.1 are illustrated in Figure 3.1. We now introduce a more systematic approach to counting the number of ways to obtain a given sum. Arrange the outcomes of the first die in ascending order, and place the outcomes of the second die above them in descending order. Summing these pairs, as shown in Figure 3.2, reveals a pattern: the lowest outcome of the first die combined with the highest outcome of the second die results in 7. Similarly, the second-lowest outcome paired with the second-highest also sums to 7, and this pattern continues. Since there are six such pairs, the probability of rolling a 7 is $\frac{6}{36}$.

Next, shift the outcomes of the second die one step to the left and repeat the process to determine the number of ways to achieve a sum of 6. Continuing this procedure systematically, we obtain the probabilities of sums ranging from 2 to 7. To find the probabilities for sums from 8 to 12, we start with Figure 3.2 and instead shift the second die to the right.

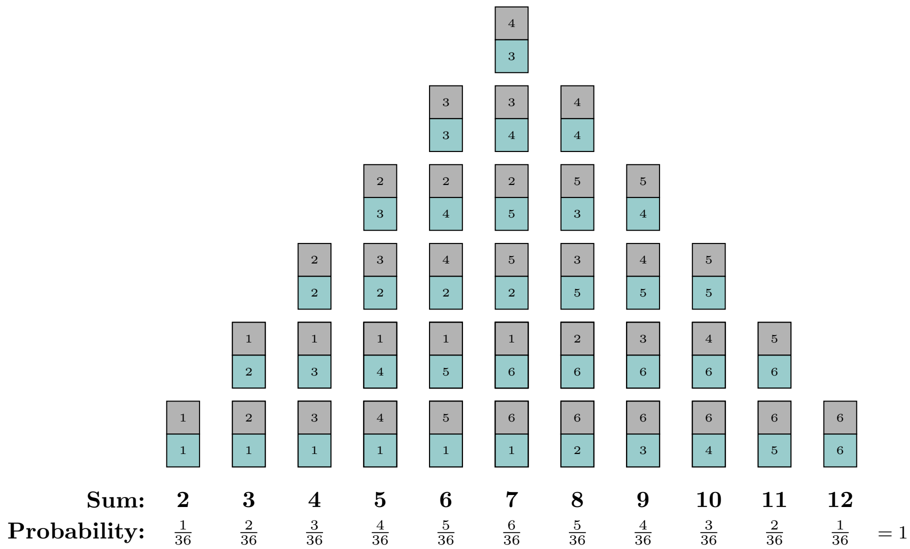

Figure 3.1: Probability of obtaining certain sum when rolling 2 dice

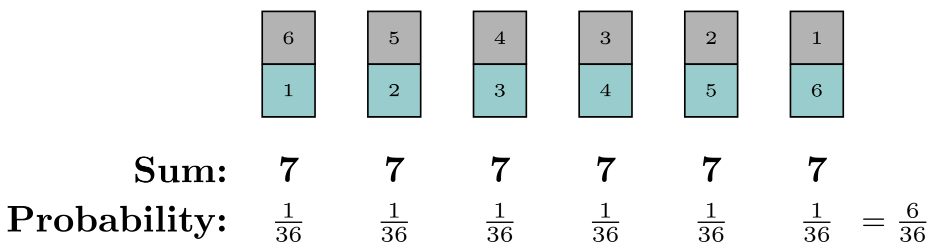

Figure 3.2: Probability of obtaining sum equal 7 when rolling 2 dice

More generally, if we have two functions and we want to understand how one influences or combines with the other, we can analyze their interaction through a systematic shifting process. By aligning one function in its natural order and the other in a reversed manner, we sum their overlapping values at each step as we slide one function over the other. This process measures how much the two functions align at each position, effectively blending them.

This idea naturally leads to the concept of *convolution*, which provides a structured way to combine two functions by considering all possible shifts and summing their interactions. In the context of our dice experiment, this method explains how the probability distribution of the sum arises from the distributions of the individual dice.

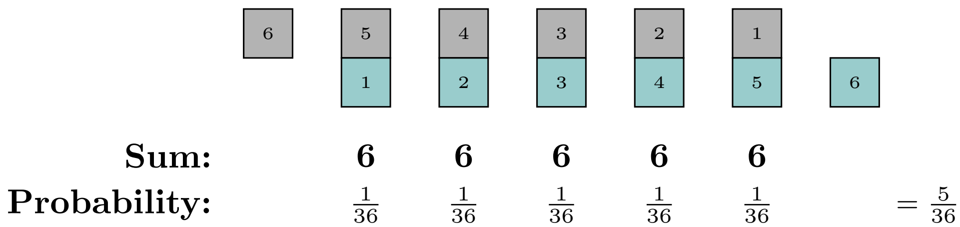

Figure 3.3: Probability of obtaining sum equal 6 when rolling 2 dice

Definition 3.1.1 (Convolution)**.** *For two measures μ and ν on $(\mathbb{R}^d, \mathcal{B})$, the convolution measure is defined as*

$$(\mu * \nu)(B) := \int \nu(B - x)\, d\mu(x)$$

for any $B \in \mathcal{B}$, where $B - x = \{y \in \mathbb{R}^d : y + x \in B\}$.

Note that the convolution operation $*$ is both associative and commutative. We will demonstrate that the probability density function of the sum of two independent random variables, $X_1 + X_2$, is given by $(f_X * f_Y)(t)$.

Proof. Let two independent random variables be X and Y with pdfs f_x and f_y.

$$\text{Prob}(X_1 + X_2 \leq t) = \iint\limits_{\{(x,y):\ x+y\leq t\}} f(x,y)\, dx\, dy = \iint\limits_{\{(x,y):\ y\leq t-x\}} f_x(x) f_y(y)\, dx\, dy.$$

Let $u = y + x,\ y = t - x \Rightarrow u = t$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{t-x} f(x,y)\, dy\, dx = \int_{-\infty}^{\infty} \int_{-\infty}^{t-x} f_x(x) f_y(u - x)\, du\, dx.$$

Notice that the inner integral is a convolution

$$f_{X+Y}(t) = \int_{-\infty}^{\infty} (f_X * f_Y)(t)\, dx.$$

Therefore, the probability density function of $X_1 + X_2$ is $(f_y * f_x)(t)$. $\square$

Similarly, if $X_1, X_2, \ldots, X_n$ are independent random variables with probability density functions $f_1, f_2, \ldots, f_n$, respectively, then the probability density function of $X_1 + X_2 + \cdots + X_n$ is $f_1 * f_2 * \cdots * f_n$.

3.2 Lindeberg Central Limit Theorem

The Central Limit Theorem tells us that for a large number of independent random variables, regardless of the shape of their individual distributions (as long as they have finite mean and variance), the distribution of their sum (or average) will approach a normal distribution (Gaussian distribution) as the number of variables increases. We will explore a version of the Central Limit Theorem that does not require the random variables to be identically distributed, specifically the Lindeberg Central Limit Theorem.

We introduce a special probability measure that measures Borel sets, called the Gaussian Measure. Remember, the probability measure of the whole space is one, here γ.

Definition 3.2.1 (Gaussian Measure on $\mathbb{R}$)**.** *A Borel measure γ on $(\mathbb{R}, \mathcal{B})$ is said to be Gaussian with mean m and variance σ^2 if*

$$\gamma((a,b]) = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b \exp\left(-\frac{1}{2\sigma^2}(x-m)^2\right) d\lambda(x).$$

Definition 3.2.2 (Gaussian Random Variable)**.** *A random variable Z from a probability space $(\Omega, \mathcal{F}, \mu)$ to $(\mathbb{R}, \mathcal{B})$ is said to be Gaussian if $\gamma := \mu \circ Z^{-1}$ is a Gaussian measure on $(\mathbb{R}, \mathcal{B})$.*

In extending the classical Central Limit Theorem beyond identically distributed settings, one encounters the necessity of organizing sequences of random variables in a meaningful and structurally coherent way. To this end, we introduce the framework of a *triangular array*: a doubly-indexed collection $\{X_{nk}\}$, where n indexes the rows and $k \in \{1, \ldots, k_n\}$ indexes the elements within each row. Each row represents a finite system of independent random variables, potentially with non-identical distributions.

Our interest lies in the behavior of the row-wise sums $Z_n = \sum_{m=1}^{k_n} X_{n,m}$ and row-wise variance $\sigma_n^2 = S_n^2 = \sum_{m=1}^{N_n} \sigma_{n,m}^2$ as n increases. In this context, a single sequence of random variables $(X_n)_{n=1}^{\infty}$ does not suffice to capture the evolving structure. The triangular array allows us to analyze convergence at the level of row-wise sums, rather than tracking individual variables in isolation.

Definition 3.2.3 (Triangular Array)**.** *A* triangular array of random variables *is a doubly-indexed collection $\{X_{n,m} : 1 \leq m \leq N_n,\ n \in \mathbb{N}\}$, where n denotes the row index and m the* *position within the nth row. The array has the form:*

$$\begin{array}{llll} X_{1,1} & & & \\ X_{2,1} & X_{2,2} & & \\ X_{3,1} & X_{3,2} & X_{3,3} & \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

where each row $(X_{n,1}, \ldots, X_{n,N_n})$ *consists of independent random variables, each with zero mean* $\mathbb{E}[X_{n,m}] = 0$ *and finite variance* $\mathbb{E}[X_{n,m}^2] < \infty$*, where the index* m *satisfies* $1 \leq m \leq N_n$.

Our triangular arrays may be composed of individual random variables that have have highly variable behavior. To guarantee that the normalized row sums still converge to a Gaussian limit, we need a condition that prevents any single variable from dominating. The Lindeberg condition does precisely this: it ensures that the influence of large deviations vanishes in the limit.

Definition 3.2.4 (Lindeberg Condition)**.** *Let* $\{X_{n,m} : 1 \leq m \leq N_n,\ n \in \mathbb{N}\}$ *be a triangular array of independent random variables such that* $\mathbb{E}[X_{n,m}] = 0$ *and* $\mathbb{E}[X_{n,m}^2] < \infty$ *for all* n, m*. We say that the array satisfies the* Lindeberg condition *if for every* $\varepsilon > 0$*,*

$$\lim_{n\to\infty} \frac{1}{S_n^2} \sum_{m=1}^{N_n} \mathbb{E}\left[X_{n,m}^2 \cdot \chi_{\{|X_{n,m}| > \varepsilon S_n\}}\right] = 0.$$

While the Lindeberg condition may seem complicated at first, its role becomes clear upon closer inspection. We begin by examining its structure to reveal the intuition behind it.

One of the assumptions in the Lindeberg–Feller Central Limit Theorem is that $\mathbb{E}[X_{n,m}] = 0$, i.e., the random variables are centered. In this case, the variance simplifies to $\mathrm{Var}[X_{n,m}] = \mathbb{E}[X_{n,m}^2]$. We now decompose this second moment into two parts:

$$\mathbb{E}[X_{n,m}^2] = \mathbb{E}\left[X_{n,m}^2 \cdot \chi_{\{|X_{n,m}| > \varepsilon S_n\}}\right] + \mathbb{E}\left[X_{n,m}^2 \cdot \chi_{\{|X_{n,m}| \leq \varepsilon S_n\}}\right],$$

where $S_n^2 = \sum_{m=1}^{N_n} \mathbb{E}[X_{n,m}^2]$ denotes the total variance of the nth row.

Dividing both sides by S_n^2, we obtain:

$$\frac{\mathbb{E}[X_{n,m}^2]}{S_n^2} = \frac{\mathbb{E}\left[X_{n,m}^2 \cdot \chi_{\{|X_{n,m}| > \varepsilon S_n\}}\right]}{S_n^2} + \frac{\mathbb{E}\left[X_{n,m}^2 \cdot \chi_{\{|X_{n,m}| \leq \varepsilon S_n\}}\right]}{S_n^2}.$$

Summing over all $m = 1, \ldots, N_n$, the left-hand side becomes:

$$\sum_{m=1}^{N_n} \frac{\mathbb{E}[X_{n,m}^2]}{S_n^2} = 1.$$

Now, taking the limit as $n \to \infty$, Lindeberg's condition tells us that the contribution from the large values—those exceeding εS_n—vanishes in the limit. Hence, we are left with:

$$1 = \lim_{n\to\infty} \sum_{m=1}^{N_n} \frac{\mathbb{E}\left[X_{n,m}^2 \cdot \chi_{\{|X_{n,m}| \le \varepsilon S_n\}}\right]}{S_n^2}.$$

Rewriting this in unnormalized form, we obtain:

$$S_n^2 = \lim_{n\to\infty} \sum_{m=1}^{N_n} \mathbb{E}\left[X_{n,m}^2 \cdot \chi_{\{|X_{n,m}| \le \varepsilon S_n\}}\right].$$

That is, in the limit $n \to \infty$, the entire variance of the sum Z_n comes from those terms whose magnitude is uniformly bounded by εS_n. The Lindeberg condition thus guarantees that large, atypical fluctuations do not dominate the sum, and that convergence to the normal distribution is governed by the collective behavior of the small, well-controlled terms.

The Lindeberg Central Limit Theorem states that if a triangular array of random variables with finite variance and zero mean satisfies Lindeberg's condition for every $\varepsilon > 0$, then the normalized sum $\frac{Z_n}{S_n}$, where $Z_n = \sum_{m=1}^{N_n} X_{n,m}$, $S_n^2 = \sum_{m=1}^{N_n} \mathbb{E}[X_{n,m}^2]$, converges in distribution to the standard normal distribution $\mathcal{N}(0, 1)$.

Theorem 3.2.1 (Lindeberg Central Limit Theorem)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a probability space, and let $\{X_n\}_{n=1}^{\infty}$ be independent (not necessarily identically distributed) random variables on $(\mathbb{R}^d, \mathcal{B})$ such that $\mathbb{E}[X_n] = 0$ and $\mathbb{E}[(X_n)^2] < \infty$. If for every $\varepsilon > 0$ Lindeberg's condition holds then*

$$\frac{Z_n}{S_n} \xrightarrow{d} \mathcal{N}(0, 1).$$

We consolidate the concepts introduced in this section through a straightforward example. Specifically, we compute the row variance S_n^2 and row standard deviation S_n for the first three rows of a triangular array and evaluate the sum$\sum_{m=1}^{N_n} \mathbb{E}\left[X_{n,m}^2 \cdot \chi_{\{|X_{n,m}| > \varepsilon S_n\}}\right]$ for the first row. Subsequently, we use the R programming language to numerically illustrate the Lindeberg Central Limit Theorem in Figure 4.2.

Example 3.2.2 (Lindenberg Condition)**.** Let the following be a triangular array composed of random variables such that columns with odd indices follow a mean-adjusted Poisson distribution, $X_{n,m} \sim \text{Bin}(5, 0.6) - 3$, and columns with even indices follow a mean-adjusted uniform distribution, $X_{n,m} \sim \mathcal{U}(0, 6) - 3$:

$$\begin{array}{llll} X_{1,1} & & & \\ X_{2,1} & X_{2,2} & & \\ X_{3,1} & X_{3,2} & X_{3,3} & \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

Since we know that $S_n^2 = \sum_{m=1}^{N_n} \sigma_{n,m}^2$ and $\text{Var}[\text{Bin}(n, p)] = n \cdot p \cdot (1 - p)$ and $\text{V}ar[\mathcal{U}(a, b)] = \frac{(a+b)^2}{12}$. Then, $S_1^2 = 1.2$, $S_2^2 = 1.2 + 3 = 4.2$, $S_3^2 = 1.2 + 3 + 1.2 = 5.4$. Now, $S_1 \approx 1.0954$, $S_2 \approx 2.0494$, $S_3 \approx 2.3238$. We know that $X \sim \text{Bin}(5, 0.6)$ can take values from 0 to 5. For simplicity, we let $\varepsilon = 1$ and check which values our indicator function sees.

$$\begin{array}{lll} |0-3| = 3 > 1.1, & |2-3| = 1 \ngtr 1.1, & |4-3| = 1 \ngtr 1.1, \\ |1-3| = 2 > 1.1, & |3-3| = 0 \ngtr 1.1, & |5-3| = 2 > 1.1. \end{array}$$

Hence, for S_1

$$\frac{1}{S_n^2} \sum_{m=1}^{N_n} \mathbb{E}\left[(X_k - \mu_k)^2 \cdot \chi_{\{|X_k - \mu_k| > \varepsilon \cdot S_n\}}\right] =$$

$$= \frac{1}{1.2} \left(P(X = 0) \cdot 9 + P(X = 1) \cdot 4 + P(X = 5) \cdot 4\right) \approx 1.54.$$

In general, if the sequence of independent random variables in question satisfies

$$\max_{k=1,\dots,n} \frac{\sigma_k^2}{S_n} \to 0, \quad \text{as } n \to \infty.$$

then Lindeberg's condition holds. Clearly, when we alternate two distributions with finite variance, Lindeberg's condition is satisfied as $\max_{k=1,\dots,n} \sigma_k^2$ for $n \geq 2$ is a constant while S_n diverges as $n \to \infty$.

4 Conclusion

The idea of measuring objects dates back at least to Archimedes. In this work, we build the theory of measure systematically from foundational principles. We begin by specifying desirable intuitive properties such as translation invariance, sigma-additivity, and consistency with the length of intervals. Ideally, measuring the power set would allow us to measure every imaginable subset. However, our first significant theorem demonstrated that such a measure cannot exist. To address this limitation, we introduced sigma-fields, structured subsets of the power set that retain our desired properties while still being sufficiently large to include all subsets of practical interest.

This brought us to the next challenge: identifying a mathematical structure that satisfies these numerous and strong conditions. To make progress, we relaxed some requirements and introduced weaker notions, such as pre-measures, outer measures, and rings. After developing intuition through examples, we proved Carathéodory's Extension Theorem, which allows us to extend pre-measures defined on rings to measures defined on sigma-fields. We showed that the collection of finite unions of semi-closed intervals forms a ring structure, which can be extended to a sigma-field using Carathéodory's Extension Theorem. The resulting measure we obtain through this process is precisely the Lebesgue measure.

The benefits of the measure-theoretic approach become evident in the second chapter, where we explore integration and lay the foundations of probability theory. We begin by introducing simple functions and measurable functions, leading to the proof of three central convergence theorems that are crucial in modern analysis. Delving deeper, we establish fundamental results such as the change of variables formula and the Fubini-Tonelli theorem, revealing the sophisticated structure underlying what are often treated as straightforward results. This powerful approach allowed us to uniformly apply definitions and theorems to both discrete and continuous random variables. In the final section, We showed how to combine probability density functions of any distributions, regardless of it being discrete or continuous. Lastly, we developed intuition for the Lindeberg Central Limit Theorem by unpacking the reasoning behind its conditions, and illustrated its practical significance through both analytical computations and R simulations that visually confirmed the convergence behavior predicted by the theorem.

Studying measure theory and its applications to probability was deeply rewarding. It gave me a greater appreciation for core results like change of variables and order of integration. Many proofs followed a beautiful, constructive path, starting with indicator functions, then simple functions, and finally measurable ones. This progression makes integration manageable by building from easily handled cases.

I was intrigued by how different areas of mathematics define "size." For instance, the Cantor set is considered "large" in topology since it has the same cardinality as the real numbers, while in measure theory, it has measure zero and is considered "small". Lastly, understanding the limitations of regular fields in handling non-measurable sets helped me see the necessity behind algebraic structures, giving me a deeper respect for how mathematical theory is constructed.

Ultimately, this project has shown that measure theory is not only a cornerstone of modern analysis and probability, but also a gateway to understanding more advanced topics across mathematics and applied fields. By building a rigorous foundation from first principles and tracing its powerful applications, I have gained both a deeper technical understanding and a broader appreciation for the elegance and utility of the theory. Whether in pure mathematics, stochastic modeling, or machine learning, the language of measure continues to shape the way we quantify uncertainty, structure complexity, and reveal the hidden patterns within the infinite.

Further Reading

The introduction was drawn from [Tao, 2011], where Dr. Tao masterfully connects various areas of mathematics, offering intuitive examples and thought-provoking exercises. His chapter on problem-solving strategies is especially insightful and hence recomended for further reading.

Definitions regardin measure in this work are mostly based on [Axler, 2020], where Dr. Axler's clear and bold presentation of definitions and proofs is highly effective. Strongly recommended when in need for a quick reference.

The proof of the existence of non-measurable sets in $\mathbb{R}$ is based on the presentations in [Landim, 2018] and [Axler, 2020], while the treatment of the Carathéodory Extension Theorem follows [Kashlak, 2024]. Dr. Kashlak also proves the uniqueness part in later videos, which are recommended for a more complete understanding of the material.

Another invaluable source was the lecture notes by Adam B. Kashlak [Kashlak, 2024], from which I adopted several definitions related to probability and drew inspiration for the presentation of the Fubini-Tonelli Theorem in Chapter 2. The only image not created by me is Figure 2.3, which is taken from [Wikipedia, date]; all other visual elements were produced by hand or generated using the LaTeX TikZ package.

The explanation of image measures was inspired by [Grossman, 2021], while the discussion of probability density functions draws from [Billingsley, 2013]. My understanding of convolution was shaped by the Stanford lecture notes on Fourier Transforms and Its Applications (EE 261) [Oppenheim and Verghese, date]. The treatment of triangular arrays and the Lindeberg Central Limit Theorem was guided by [Breheny, date], and the explanation of the Lindeberg condition was influenced by [Rós, 2005].

Another possible direction for this project could have been the study of stochastic processes, particularly Brownian motion. The paths of Brownian motion, denoted $B_t(\omega)$, are random functions of time t that are continuous everywhere but differentiable nowhere. Their extreme irregularity makes them incompatible with classical Riemann integration. However, measure theory, through Lebesgue integration, provides the rigorous tools needed to handle such functions. It ensures their measurability and enables precise definitions of expectations, variances, and other key probabilistic concepts.

Another direction could have focused on the theory behind machine learning. Measure theory provides the mathematical foundation for probability, integration, and expectation. These concepts are central to how machine learning models are trained, evaluated, and interpreted. A project in this area might have explored applications such as explainable AI.

Appendix 1

A *relation* is a way of associating an element of a set to an element of another set (or itself). We define relations more formally:

Definition 4.0.1 (Relation)**.** *Let A and B be two sets. A relation R from A to B is a subset of the Cartesian product $A \times B$. That is,*

$$R \subseteq A \times B.$$

In other terms, R consists of some ordered pairs (a, b) where $a \in A$ and $b \in B$. If $(a, b) \in R$, we write aRb.

Example 4.0.1 (Relation I)**.** Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. We define some relation $R \subseteq A \times B$ as follows:

$$R = \{(1, a), (2, b)\}.$$

This means that 1 is related to a, 2 is related to b, and 3 is not related to any element of B under this relation.

Example 4.0.2 (Relation II)**.** Inequalities on $\mathbb{R}$ are relations. For instance, "is less than" can be written as a subset of $\mathbb{R} \times \mathbb{R}$:

$$R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x < y\}.$$

Definition 4.0.2 (Basic Properties of Relations)**.** *Let A be a set. A relation R from A to itself is:*

- ***reflexive*** *if aRa for all $a \in A$,*
- ***symmetric*** *if $aRb \Rightarrow bRa$ for all $a, b \in A$,*
- ***antisymmetric*** *if $(aRb$ and $bRa) \Rightarrow a = b$ for all $a, b \in A$,*
- ***transitive*** *if $(aRb$ and $bRc) \Rightarrow aRc$ for all $a, b, c \in A$.*

Definition 4.0.3 (Equivalence Relation)**.** *An equivalence relation on a set A is a binary relation R on A that is reflexive, symmetric, and transitive. We commonly choose to denote a relation by $\sim$ if it is an equivalence relation.*

Example 4.0.3 (Equivalence Relation)**.** For any set S, equality is an equivalence relation.

Equivalence relations are useful because they allow us to partition a set into *equivalence classes*. We give the appropriate definition and proof:

Definition 4.0.4 (Equivalence Class)**.** *Let S be a set and $\sim$ be an equivalence relation on S. Then, for all $a \in S$, the equivalence class of a is*

$$[a] = \{x \in S : x \sim a\}.$$

Theorem 4.0.4 (Partitioning Sets)**.** *Let S be a set and $\sim$ be an equivalence relation on S. The equivalence classes of $\sim$ form a partition of S.*

Proof. See that any element of S belongs to an equivalence class. For any $x \in S$, we have $x \in [x]$ since $x \sim x$ by the reflexive property.

Furthermore, any two equivalences classes are either identical or disjoint. Let $a, b \in S$. Suppose that $[a] \cap [b] \neq \emptyset$. Then, there exists some $c \in [a] \cap [b]$. We have $c \sim a$ and $c \sim b$. By the reflexive property, we also have $a \sim c$. Notice that for any $x \in [a]$, we have that $x \sim a$, $a \sim c$, and $c \sim b$. Applying the transitivity property twice tells us that $x \in [b]$. Thus, any $x \in [a]$ satisfies $x \in [b]$, meaning that $[a] = [b]$. Otherwise, we have $[a] \cap [b] = \emptyset$.

Since any element of S belongs to an equivalence class and any two equivalence classes are either identical or disjoint, it follows that the equivalence classes of $\sim$ form a partition of S. $\square$

Example 4.0.5 (Partition of Z)**.** Congruence modulo n is defined on $\mathbb{Z}$ as follows:

$$a \equiv b \bmod n \Leftrightarrow \frac{a-b}{n} \in \mathbb{Z}.$$

This is an equivalence relation with equivalence classes determined by the remainder of an integer when divided by n. As such, we have the equivalence classes $[0], [1], \ldots, [n-1]$.

Appendix 2

Proof of Monotone Class Theorem. In this proof, we will show that $\sigma(\mathcal{A}) \subseteq m(\mathcal{A})$, where $m(\mathcal{A})$ is the smallest monotone class that contains $\mathcal{A}$. In other words, $m(\mathcal{A})$ is the intersection of all monotone classes that contain $\mathcal{A}$. This is done by showing that $m(\mathcal{A})$ is a field, and thus a σ-field, and therefore contains $\sigma(\mathcal{A})$ since $\sigma(\mathcal{A})$ is minimal.

Step (1) Show that $m(\mathcal{A})$ is closed under taking complements. Let $\mathcal{F} = \{A \in m(\mathcal{A}) : A^c \in m(\mathcal{A})\}$. Now, $\mathcal{A} \in \mathcal{F}$ since $\mathcal{A}$ is closed under complements. That is, $\forall A \in \mathcal{A},\ A^c = \mathbb{X} \setminus A \in \mathcal{A}$.

Furthermore, for a sequence $\{A_i\}_{i=1}^{\infty}$ such that $A_i \in \mathcal{F}$ and $A_i \uparrow A = \bigcup_{i=1}^{\infty} A_i$, we have $A_i^c \in \mathcal{F}$ and $A_i^c \downarrow A^c = \bigcap_{i=1}^{\infty} A_i^c = \left(\bigcup_{i=1}^{\infty} A_i\right)^c$. This could be visualized with Figure 4.1. Therefore, $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$. Thus, $\mathcal{F}$ is monotone and $\mathcal{F} \subseteq m(\mathcal{A})$. Doing the same for $A_i \downarrow A$ shows that $\mathcal{F} = m(\mathcal{A})$ by minimality of $m(\mathcal{A})$, and therefore, $m(\mathcal{A})$ is closed under complementation.

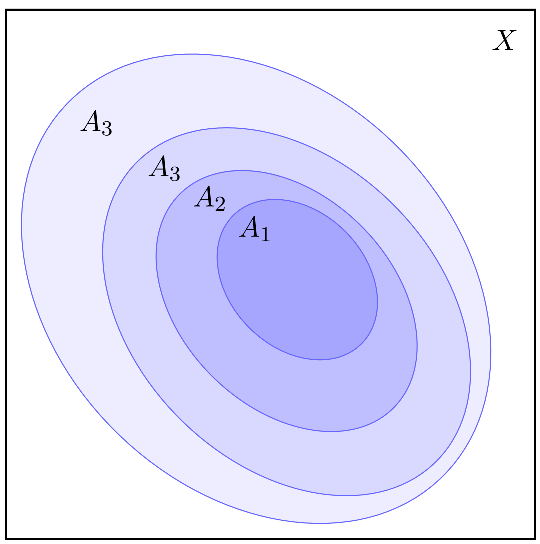

Figure 4.1: Monotonically increasing sets

Step (2) is to show that $m(\mathcal{A})$ is closed under finite unions. Let

$$G_1 = \{A \in m(\mathcal{A}) : A \cup B \in m(\mathcal{A}) \text{ for all } B \in \mathcal{A}\}.$$

As with $\mathcal{F}$ from the previous paragraph, we note that $\mathcal{A} \in G_1$. Also, G_1 is monotone, since

$(\bigcup_{i=1}^{\infty} A_i) \cup B = \bigcup_{i=1}^{\infty}(A_i \cup B) \in m(\mathcal{A})$ and $(\bigcap_{i=1}^{\infty} A_i) \cup B = \bigcap_{i=1}^{\infty}(A_i \cup B) \in m(\mathcal{A})$. Thus, by minimality, $m(\mathcal{A}) = G_1$, and hence $m(\mathcal{A})$ is closed under finite unions with elements of $\mathcal{A}$. Now, let

$$G_2 = \{B \in m(\mathcal{A}) : A \cup B \in m(\mathcal{A}) \text{ for all } A \in m(\mathcal{A})\}.$$

Once again, $\mathcal{A} \subseteq G_2$, since if $B \in \mathcal{A}$, then $A \cup B \in m(\mathcal{A})$ for any $A \in m(\mathcal{A})$, by the argument involving G_1. Hence, G_2 is monotone. Therefore, $G_2 = G_1 = m(\mathcal{A})$, which implies that $m(\mathcal{A})$ is closed under finite unions. As a result, $m(\mathcal{A})$ is a field, and consequently, it is a σ-field. Hence, by minimality, $\sigma(\mathcal{A}) \subseteq m(\mathcal{A})$. $\square$

```r
n <- 50
m <- 10
var_binom <- 5 * 0.6 * (1 - 0.6)
var_uniform <- (4 - 0)^2 / 12
Z_n <- c()  # Creating empty vector
S_n <- c()
N <- c()

for (i in 1:n) {
  for (j in 1:m) {
    if (j %% 2 == 1) {  # Odd indices: Binomial distribution
      a <- rbinom(1, 5, 0.6) - 3
      Z_n <- c(Z_n, a)
      S_n <- c(S_n, var_binom)
    } else {  # Even indices: Uniform distribution
      b <- runif(1, min = 0, max = 6) - 3
      Z_n <- c(Z_n, b)
      S_n <- c(S_n, var_uniform)
    }
  }
  N <- c(N, (sum(Z_n) / sqrt(sum(S_n))))
  Z_n <- c()  # Resetting empty vector
  S_n <- c()
}

hist(N, breaks = n, freq = FALSE,
     main = "Histogram of Normalised Z_10",
     xlab = "Values", col = "lightblue")

lines(seq(min(N), max(N), by = 1/n),
      dnorm(seq(min(N), max(N), by = 1/n), mean = 0, sd = 1),
      col = "blue")

legend("topright",
       legend = c("Normalised Z_10", "N(0,1)"),
       col = c("skyblue", "blue"),
       lty = c(1, 1),
       cex = 1.1)
```

Listing 4.1: R Code for Normalised Z_{10}

```r
n<-250
m<-50
var_binom<-5*0.6*(1-0.6)
var_uniform<-(4-0)**2/12
Z_n<-c() #creating empty vector
S_n<-c()
N<-c()
for (i in 1:n) {
  for (i in 1:m) {
    if (i %% 2 == 1) {  # Odd indices: Binomial distribution
      a<-rbinom(1,5,0.6)-3
      Z_n<-c(Z_n,a)
      S_n<-c(S_n,var_binom)
    } else {  # Even indices: Poisson distribution
      b<-runif(1, min = 0, max = 6)-3
      Z_n<-c(Z_n,b)
      S_n<-c(S_n,var_uniform)
    }
  }
  #print(sum(Z_n)/sum(S_n))
  N<-c(N,(sum(Z_n)/sqrt(sum(S_n))))
  Z_n<-c() #creating empty vector
  S_n<-c()
}
hist(N, breaks = n, freq = F, main = "Histogram of normalised Z_50", xlab =
    "Values", col = "lightblue")
lines(seq(min(N),max(N),by=1/n),dnorm(seq(min(N),max(N),by=1/n),mean=0,sd
   =1),col="blue")
legend("topright",
       legend = c("Normalised Z_50", "N(0,1)"),
       col = c("skyblue", "blue"),
       lty = c(1, 1),  # Use 1 for solid lines
       cex = 1.1)
```

Listing 4.2: R Code for Normalised Z_{50}

```r
 n<-400
m<-500
var_binom<-5*0.6*(1-0.6)
var_uniform<-(4-0)**2/12
Z_n<-c() #creating empty vector
S_n<-c()
N<-c()
for (i in 1:n) {
  for (i in 1:m) {
    if (i %% 2 == 1) {  # Odd indices: Binomial distribution
      a<-rbinom(1,5,0.6)-3
      Z_n<-c(Z_n,a)
      S_n<-c(S_n,var_binom)
    } else {  # Even indices: Poisson distribution
      b<-runif(1, min = 0, max = 4)-2
      Z_n<-c(Z_n,b)
      S_n<-c(S_n,var_uniform)
    }
  }
  #print(sum(Z_n)/sum(S_n))
  N<-c(N,(sum(Z_n)/sqrt(sum(S_n))))
  Z_n<-c() #creating empty vector
  S_n<-c()
}
hist(N, breaks = n, freq = F, main = "Histogram of normalised Z_500", xlab
    = "Values", col = "lightblue")
lines(seq(min(N),max(N),by=1/n),dnorm(seq(min(N),max(N),by=1/n),mean=0,sd
    =1),col="blue")
legend("topright",
       legend = c("Normalised Z_500", "N(0,1)"),
       col = c("skyblue", "blue"),
       lty = c(1, 1),  # Use 1 for solid lines
       cex = 1.1)
```

Listing 4.3: R Code for Normalised Z_{500}

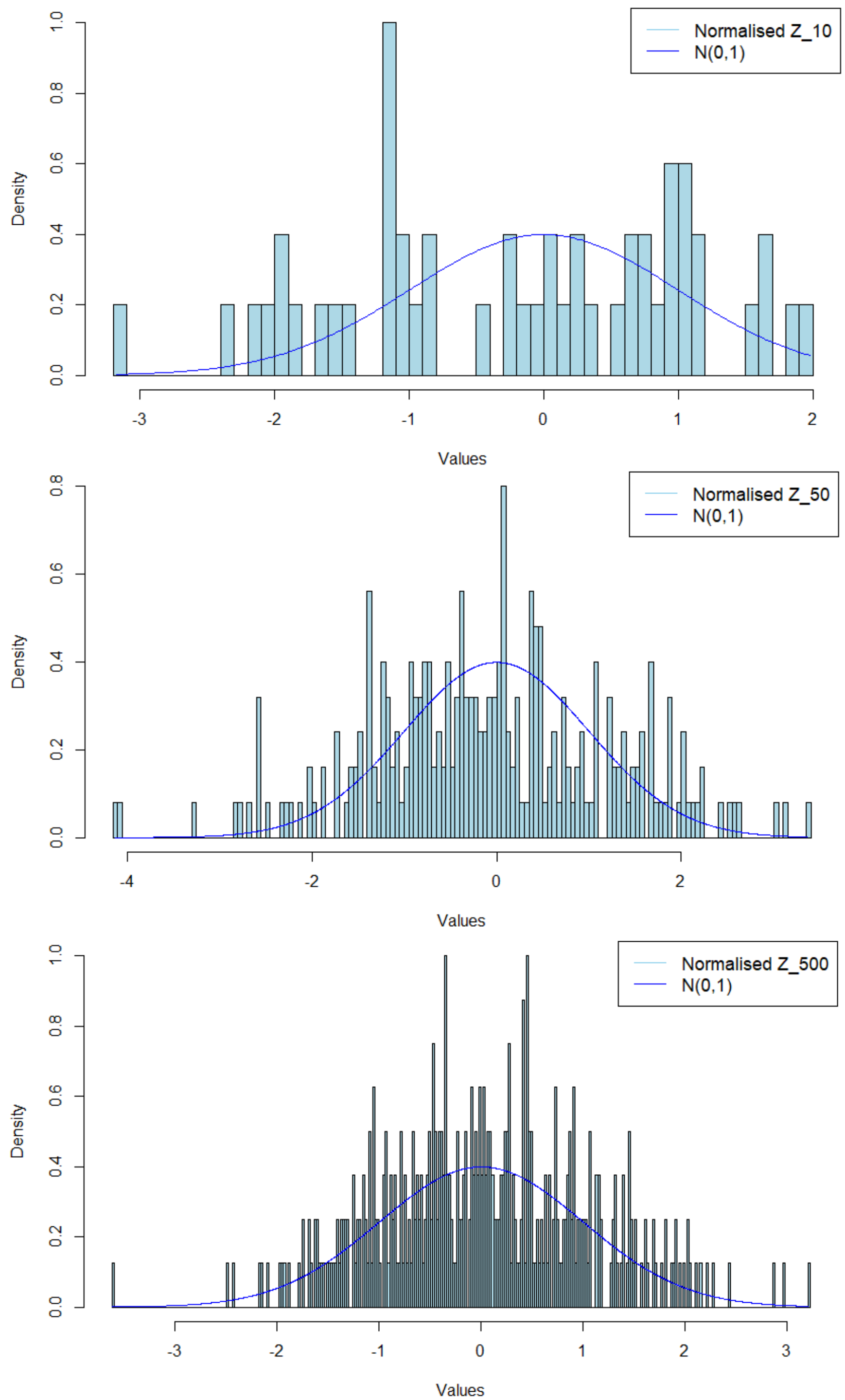

Figure 4.2: Convergence of $X_{n,m} \sim \mathrm{Bin}(5, 0.6) - 3$ and $X_{n,m} \sim \mathcal{U}(0, 6) - 3$ to $\mathcal{N}(0, 1)$

Bibliography

[Axler, 2020] Axler, S. (2020). *Measure, integration & real analysis*. Springer Nature.

[Billingsley, 2013] Billingsley, P. (2013). *Convergence of probability measures*. John Wiley & Sons.

[Breheny, date] Breheny, P. (no date). Lindeberg-feller central limit theorem. Accessed: 4 April 2025.

[Dudley, 2018] Dudley, R. M. (2018). *Real analysis and probability*. Chapman and Hall/CRC.

[Grossman, 2021] Grossman, D. (2021). Lecture 15: Sums of random variables. YouTube video. Accessed: 4 April 2025.

[Kashlak, 2024] Kashlak, A. B. (Winter 2024). Teaching. Accessed: 4 April 2025.

[Landim, 2018] Landim, C. (4 May 2018). Lecture 01: Introduction: a non-measurable set. YouTube Channel: Instituto de Matemática Pura e Aplicada.

[Oppenheim and Verghese, date] Oppenheim, A. V. and Verghese, G. C. (no date). Signals, systems and inference. PDF, Stanford Engineering Everywhere. Accessed: 4 April 2025.

[Rós, 2005] Rós, S. (2005). Glósóli. YouTube video. Accessed: 4 April 2025.

[Tao, 2011] Tao, T. (2011). An introduction to measure theory. American Mathematical Society.

[Wikipedia, date] Wikipedia (no date). Limit inferior and limit superior. Accessed: 4 April 2025.